



WELCOME!

to The Complete Data Management Masterclass

What will you learn in this course?

Course Modules

Module 1 - The Basics of Data Management

Module 2 - Data Ethics

Module 3 - Data Governance

Module 4 - Data Architecture

Module 5 - Data Modeling and Design

Module 6 - Data Storage and Operations

Module 7 - Data Security

Module 8 - Data Integration & Interoperability

Module 9 - Document and Content Management

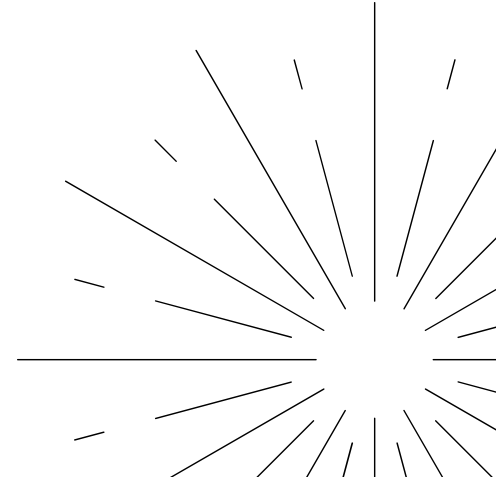
Module 10 - Master & Reference Data Management

Module 11 - Data Warehousing & Business Intelligence

Module 12 - Metadata Management

Module 13 - Data Quality

Module 14 - Big Data & Data Science



Why take this course?

Reason 1 - We cover every Data Management subject area

Reason 2 - Real examples of use cases

Reason 3 - CDMP prep

Reason 4 - Tips and tricks from experience

Reason 5 - Answers most questions that you can expect in interviews

Reason 6 - Gain confidence in attending any data meeting

Reason 7 - Additional resources provided for further reading



Download the course resources

Main resource - PDF Presentation file with all lessons

Other resources - additional files/resources provided in various lessons

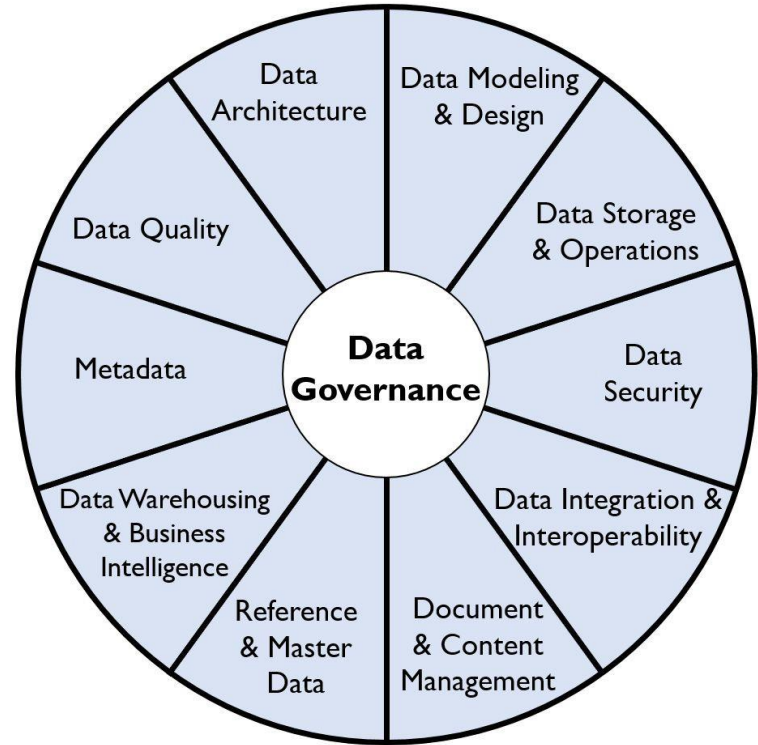
Module 1 - The Basics

What is Data Management?

‘Data management is the practice of collecting, organizing, protecting and storing an organization’s data so it can be analyzed for business decisions.’

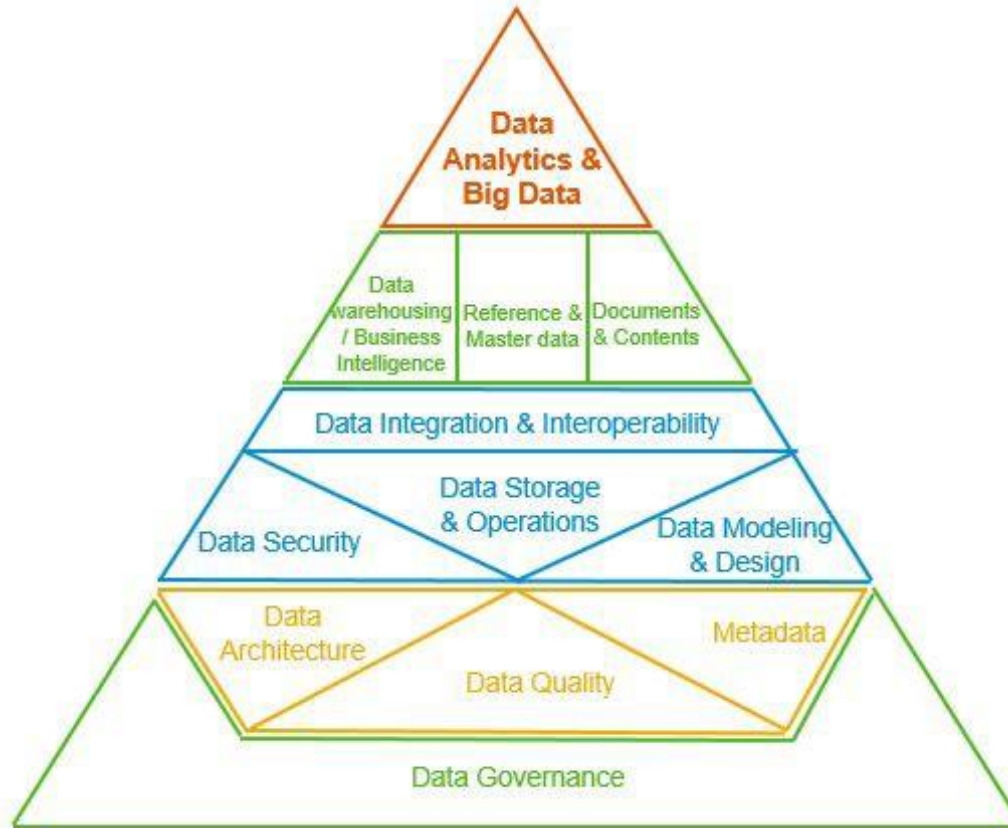
Data Management Subject Areas

1. Data Governance
2. Data Modeling & Design
3. Data Storage & Operations
4. Data Security
5. Data Integration & Interoperability
6. Document & Content Management
7. Reference & Master Data
8. Data Warehousing & Business Intelligence
9. Metadata
10. Data Quality
11. Data Architecture



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DMBOK pyramid by Peter Aiken



Source: Peter Aiken developed the DMBOK pyramid

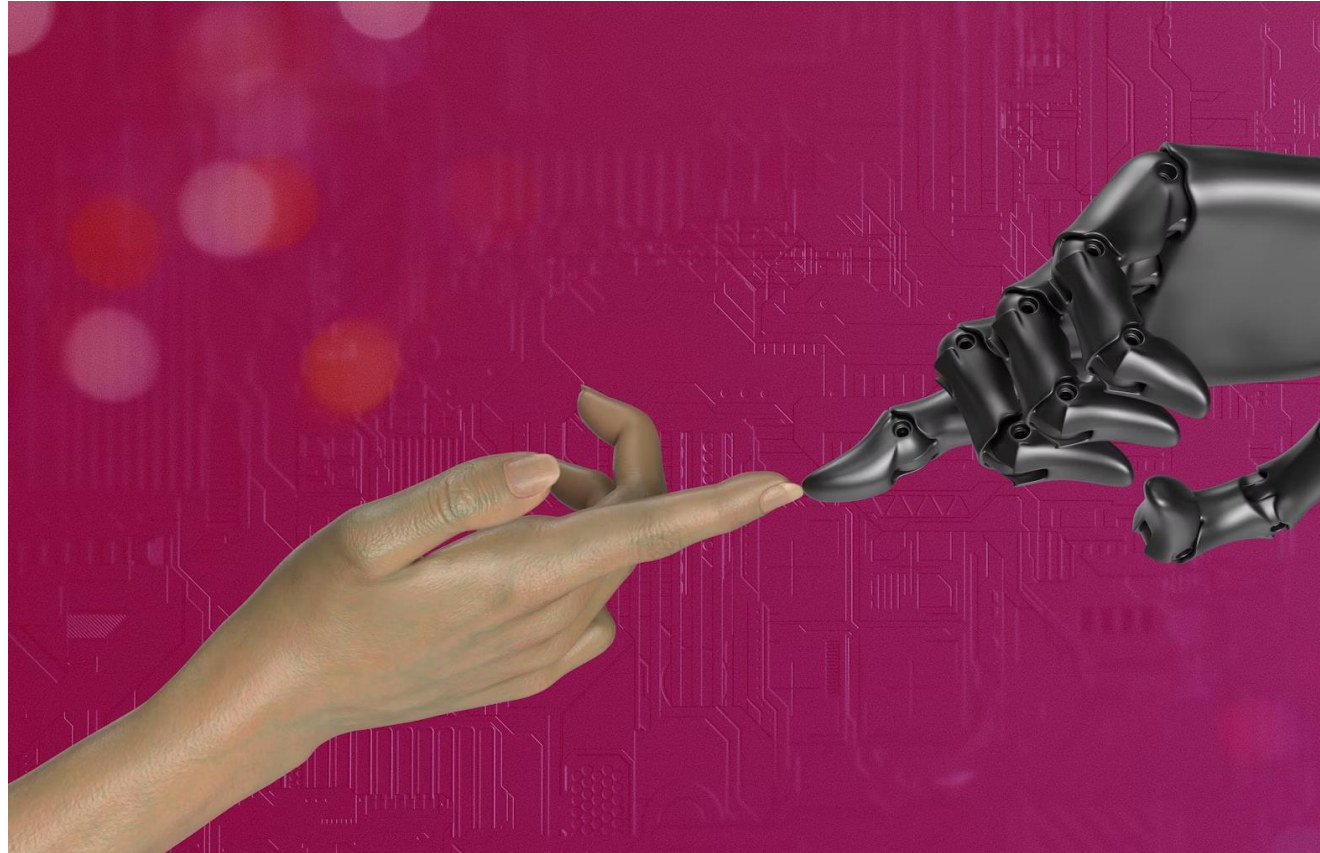
Module 2 - Data Ethics

Definition of Data Ethics

Data ethics refers to the set of moral principles that guide how data is collected, stored, analyzed, and shared.

It addresses various ethical considerations:

- Privacy
- Consent
- Transparency
- Fairness
- Accountability



Facebook and Cambridge Analytica

In 2018, the Cambridge Analytica scandal emerged as one of the most significant examples of poor data ethics. The UK-based political consulting firm gained unauthorized access to the personal data of approximately 87 million Facebook users. This data was initially collected for academic research but was misused for political profiling and targeted advertising during elections.

Consequences of the Scandal

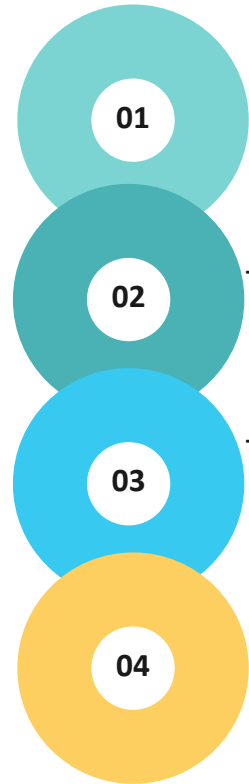
- Financial Impact: \$36 billion in Facebook's market value. Additionally regulatory fines totaling nearly \$6 billion from FTC.
- Reputational Damage
- Regulatory Scrutiny

Ethical Violations Highlighted

- Lack of Consent
- Data Misuse
- Transparency Issues



GOALS OF DATA ETHICS

- 
- 01 To define ethical handling of data in the organization
 - 02 To educate staff on the organization risks of improper data handling
 - 03 To change/install preferred culture and behaviours on handling data
 - 04 To monitor regulatory environment, measure, monitor and adjust organization approaches for ethics in data

Risk of Unethical Data Handling

What	What is it
Timing	It is possible to lie through omission or inclusion of certain data points in a report or activity based on timing.
Misleading Visualizations	Charts and graphs can be used to present data in a misleading manner.
Unclear Definitions or Invalid Comparisons	Statistical 'smoothing' of numbers over a period could completely change perception of the number.
Bias	In statistics, bias refers to deviations from expected values. These are often introduced through systematic errors in sampling or data selection.
Transforming and Integrating Data	Data integration presents ethical challenges because data is changed as it moves from system to system.
Obfuscation / Redaction of Data	Obfuscating or redacting data is the practice of making information anonymous, or removing sensitive information.

Common types of bias when working with data

Confirmation Bias

Anchor Bias

Outliers Bias

Selection Bias

**Rush-to-Solve
Bias**

Availability Bias

Confirmation Bias

Example:

- Internet searching - "Why Schumacher is the greatest Formula 1 driver"

How to avoid Confirmation Bias:

1. Understand your assumptions and beliefs before you proceed with the data analysis
2. Don't jump to conclusions due to your knowledge
3. Compare your beliefs to the findings



Outliers Bias

Example:

Averages of a KPI that do not represent the real picture

You have 5 Sales in October

Sale 1 = \$5

Sale 2 = \$7

Sale 3 = \$6

Sale 4 = \$52

Sale 5 = \$7

Average = \$15.40

How to avoid Confirmation Bias:

1. Do not fully trust averages
2. Understand the distribution of data
3. Look into the outliers and understand their impact
4. Decide what measurement will make most sense



Selection Bias

Example:

Leadership deciding on specific dataset to be analyzed.
Tom (General Manager UK) wants you to analyze the potential of expanding into a new sales territory in the UK by only analyzing a specific set of customers.

How to avoid Selection Bias:

1. User random selection of data if you cannot process all data
2. Make sure the data provides a good representation for the decisions that will need to be made
3. Do not base your data selection based on opinion of people



Rush-to-Solve Bias

Example:

Leadership wants to make a quick decision and decides to make conclusions based on limited set of data.

For example making Sales Targets for next year only based on finalized numbers for the first half of current year

How to avoid Rush-to-Solve Bias:

1. Think about the worst case scenario if wrong decision is made due to rush-to-solve attitude
2. What is the benefit of making a decision faster compared to the worst case scenario?
3. Choose what is more important



Availability Bias

Example:

News reporting multiple plane crashes over a period of a week. This leads to the audience start perceiving that plane crashes are more common than they actually are. Not based on actual statistical analysis.

A team makes business decisions based on easily available data, ignoring the data that will be harder or more time consuming to gather.

How to avoid Availability Bias

Three questions to ask yourself

1. Were you provided all the data needed?
2. Can you really afford to make business decision on limited data?
3. Can you wait to make a decision until you gather all the information?



Anchor Bias

Example:

Purchasing an item after doing research in 2 stores and believing you made a great deal since you bought it from the second store for \$30 less.

Later you find out that you could have bought the item for \$80 cheaper.

How to avoid Anchor Bias

1. Don't rush to conclusions based on the initial findings in a dataset
2. Take your time to understand the data
3. Do comprehensive analysis of the data before you make any final conclusions



What Data Ethics actually involves

6 Key Activities

1. Review Data-Handling Practices

2. Identify Principles, Practices, and Risk Factors

3. Create Ethical Data Handling Strategy

4. Address Practices Gaps

5. Communicate and Educate Staff

6. Monitor and Maintain Alignment

Deliverables of Data Ethics

8 Key Deliverables

1. Current Practices and Gaps

2. Ethical Data Handling Strategy

3. Communication Plan

4. Ethics Training Program

5. Ethical Corporate Statements on Data

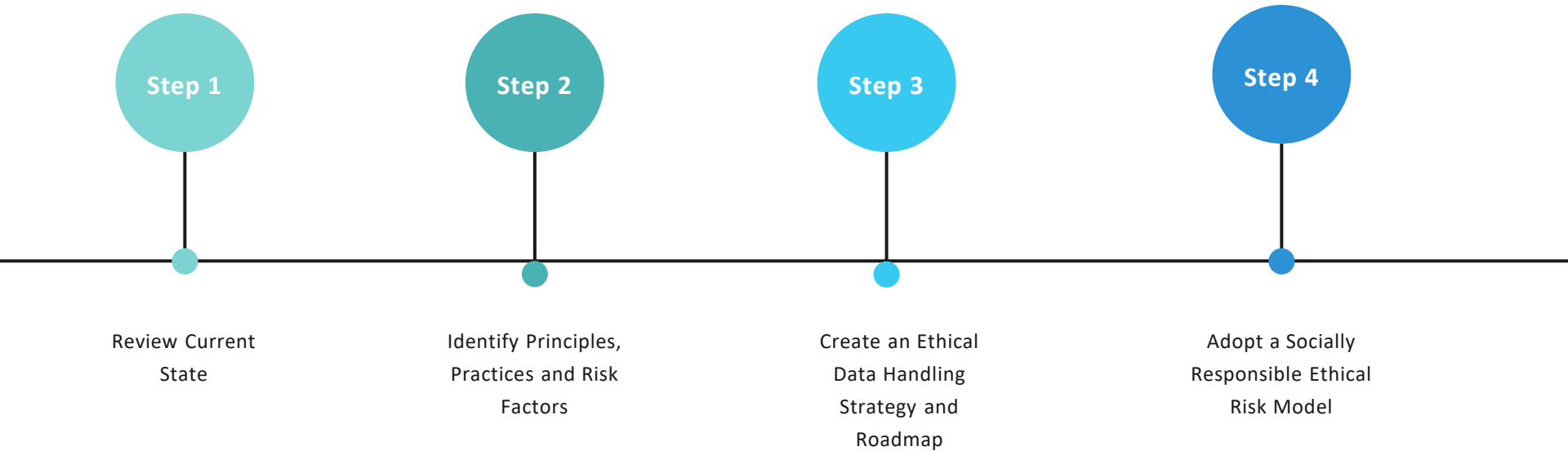
6. Aligned Incentives, KPIs and targets

7. Updated Data Policies

8. Ethical Data Handling Reporting

Establishing Ethical Data Culture

Step by Step



Module 3 - Data Governance

What is Data Governance?

- **Rules, Processes and Accountability** that allow the organization to better manage the availability, usability, security and integrity of the corporate data sources.
- **Tip - Think about it as bringing data under control and keeping it secure and consistent.**

7 reasons why you need Data Governance

1. Secure your data
2. Ensure compliance with regulations and data privacy laws
3. Improve the data quality
4. Avoid inconsistent data silos
5. Improve trust in the data
6. Better decision making
7. Improve efficiency



Regulations and Data

Amazon: \$886 million

WhatsApp - \$267 million

Home Depot: \$200 million

Uber: \$148 million

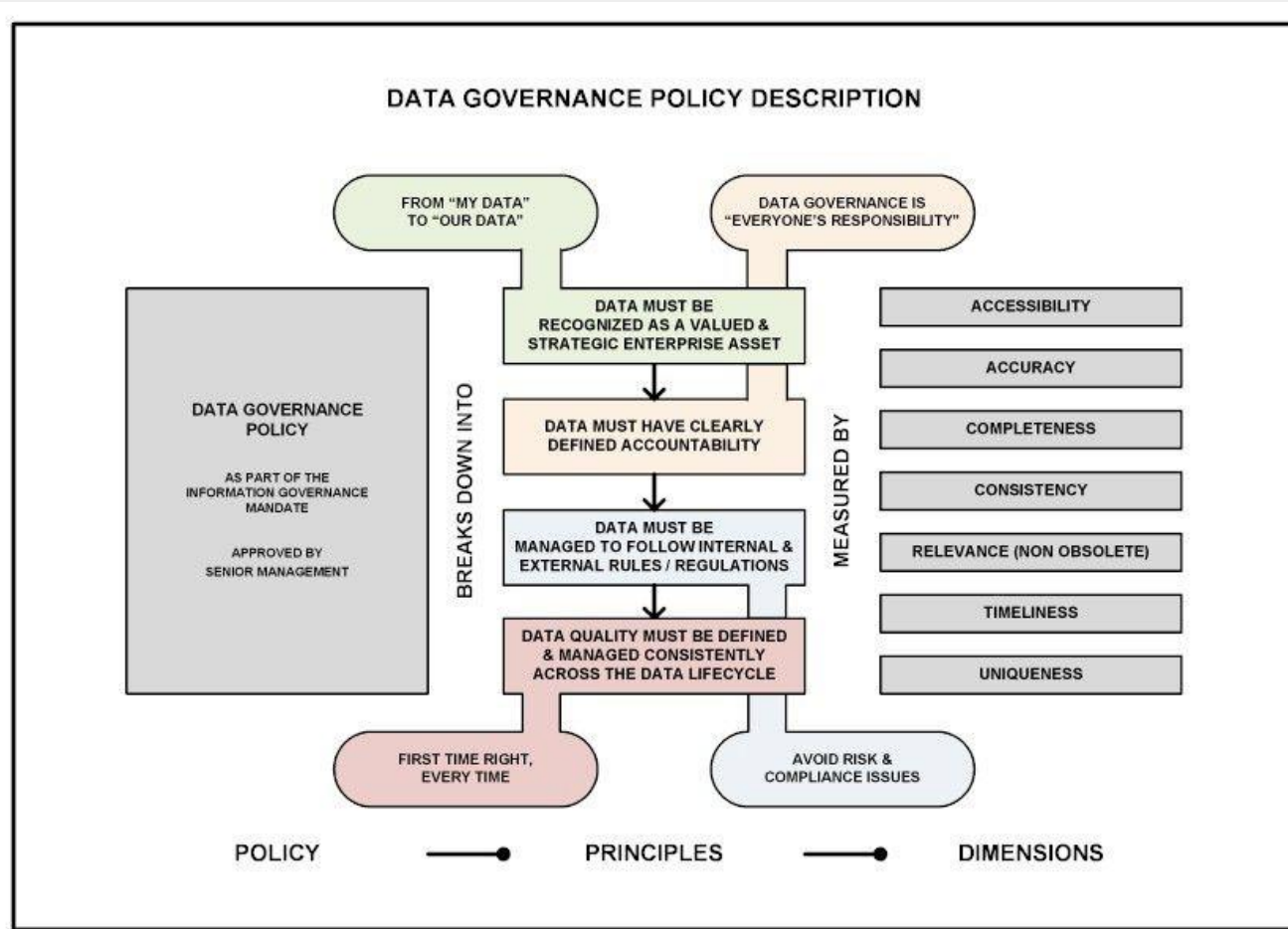
Yahoo: \$85 million

Capital One: \$80 million

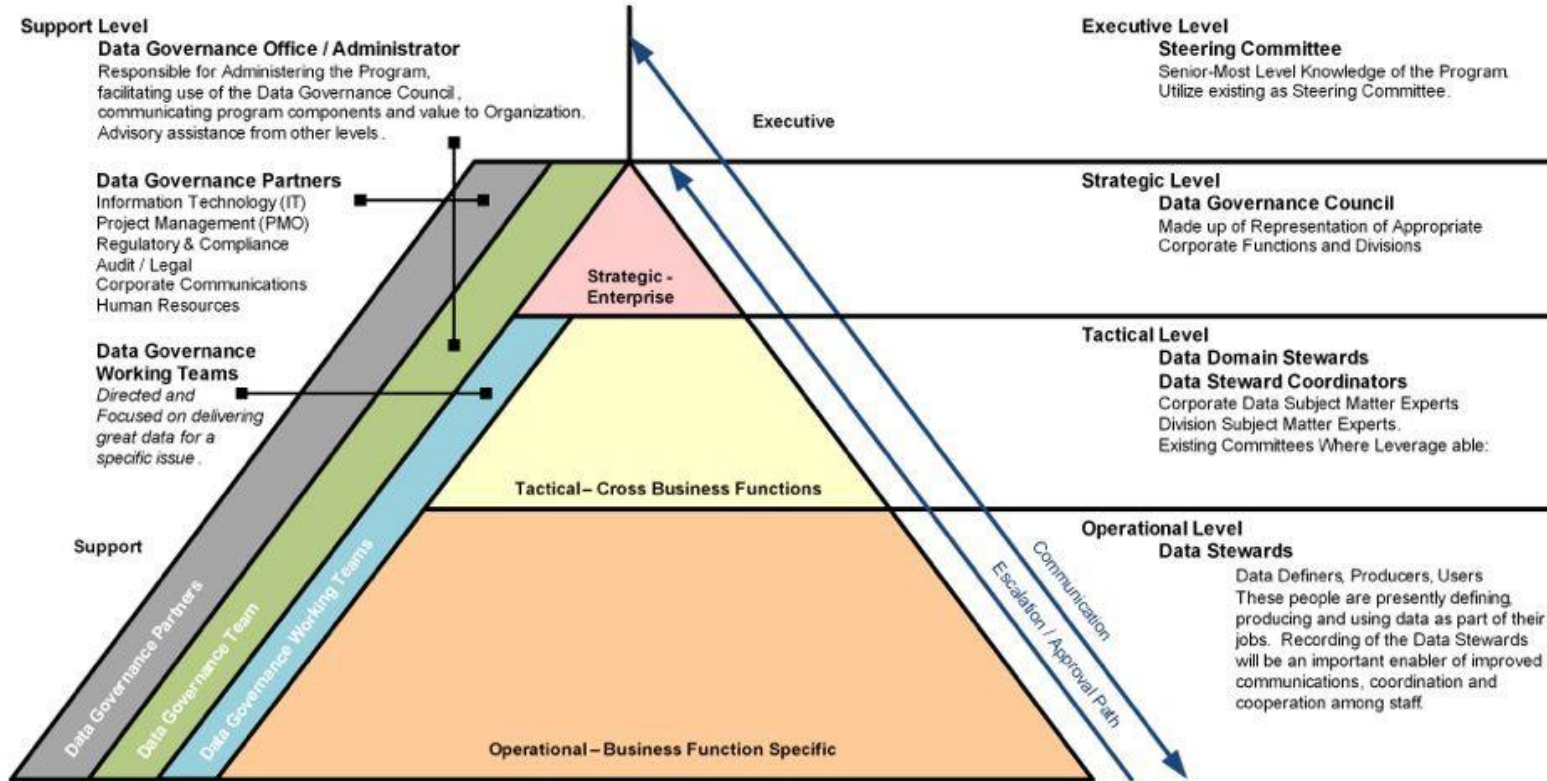


*The U.S. relies on a [“combination of legislation, regulation and self-regulation”](#) rather than government intervention alone. There are approximately 20 industry- or sector-specific federal laws, and more than 100 privacy laws at the state level (in fact, there are 25 privacy-related laws in California alone).

Data Governance Core Principles



Governance Roles and Responsibilities



Module 4 - Data Architecture

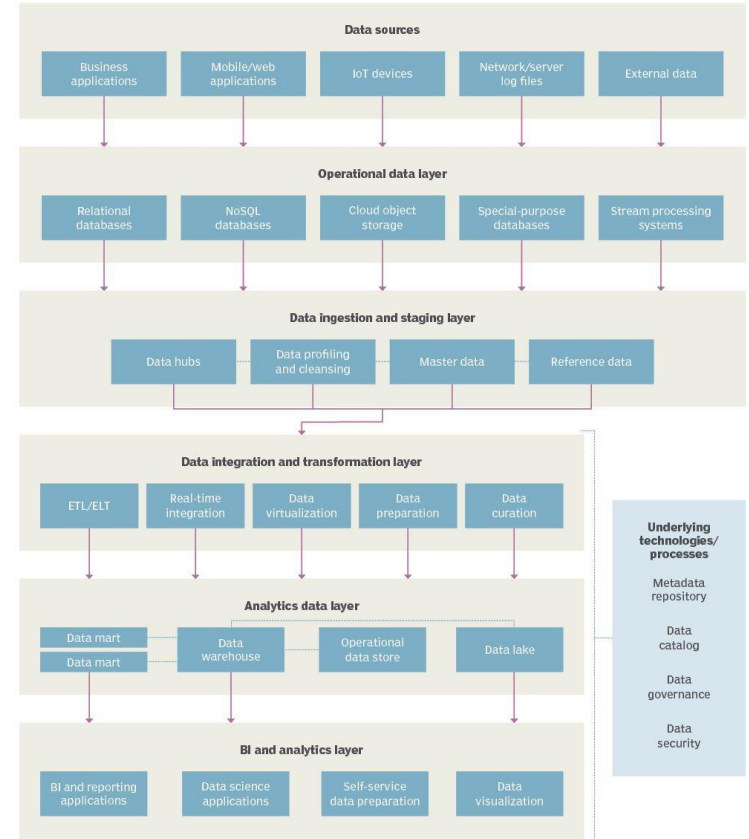
What is Data Architecture

‘**Data architecture** is a set of rules, policies, standards and models that govern and define the type of data collected and how it is used, stored, managed and integrated within an organization and its database systems. It provides a formal approach to creating and managing the flow of data and how it is processed across an organization’s IT systems and applications”

Data Architecture Example

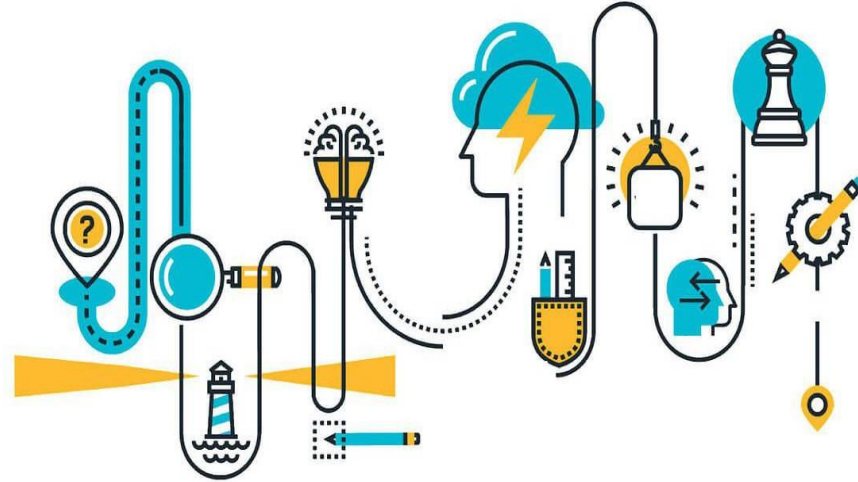
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Sample data architecture diagram



Data Architecture Principles

1. *Data is a shared asset*
2. *Users require adequate access to data*
3. *Security is essential*
4. *Common vocabularies ensure common understanding*
5. *Data should be curated*
6. *Data flows should be optimized for agility*



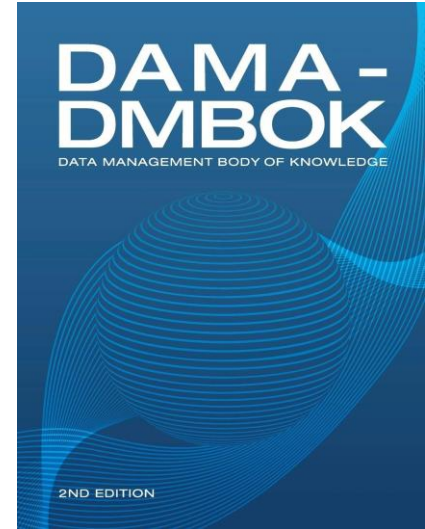
Data Architecture components

1. *Data pipelines*
2. *Cloud storage*
3. *Cloud computing*
4. *AI and ML models*
5. *Data streaming*
6. *Container orchestration*
7. *Real-time analytics*



Data Architecture frameworks

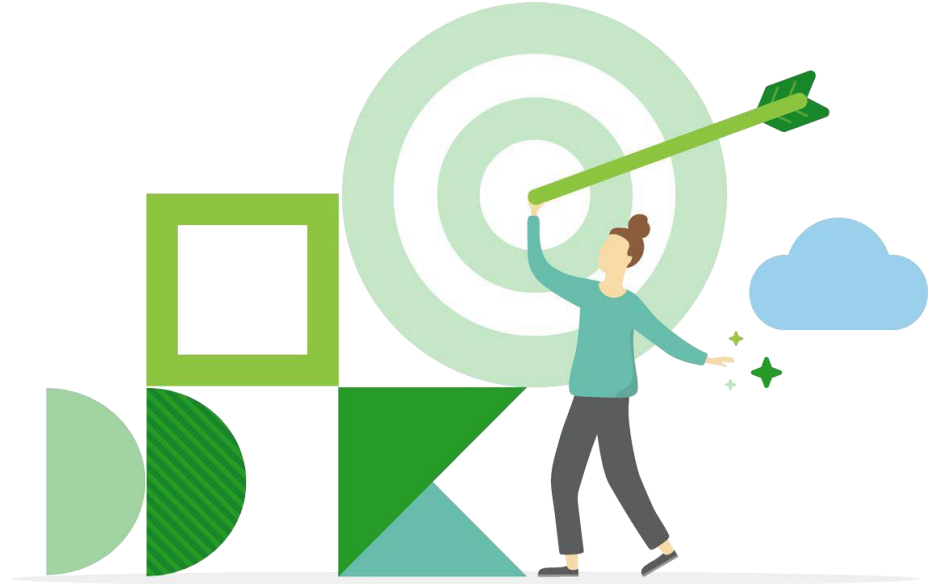
1. *DAMA-DMBOK-2*
2. *Zachman Framework for Enterprise Architecture*
3. *The Open Group Architecture Framework (TOGAF)*



TOGAF®

Data Architecture best practices

1. *Cloud-native*
2. *Robust and scalable data pipelines*
3. *Seamless data integration*
4. *Real-time data enablement*
5. *Decoupled and extensible*
6. *Domain-driven*
7. *Balanced*



Data Architecture Roles

Data Architect!!!

Data Modeler

Data Integration Developer

Data Engineer



Module 5 - Data Modeling and Design

What is Data Modeling

‘**Data modeling** is the process of creating a visual representation of either a whole information system or parts of it to communicate connections between data points and structure.

The goal is to illustrate the types of data used and stored within the system, the relationships among these data types, the ways the data can be grouped and organized and its formats and attributes.”

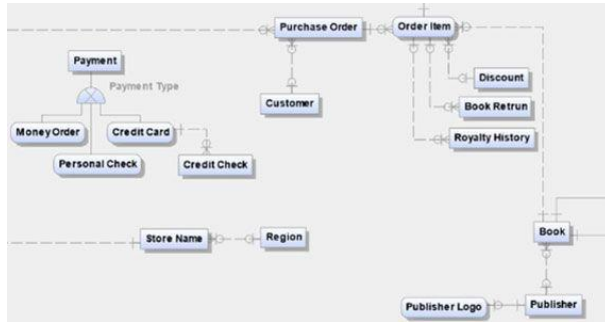
Data Modeling vs Data Architecture

5 Key Differences

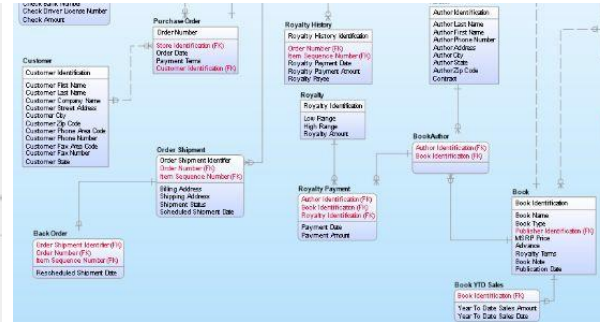
Data Modeling	Data Architecture
Focuses on the representation of the data	Focuses on what tools and platforms to use for storing and analyzing data
Focus on accuracy of the data	Focus is on the infrastructure housing the data
Focus on reliability of the data	Focus on keeping the data safe
Representation of reality	Framework of systems and logistics
Represents a limited set of business concepts	Covers the data infrastructure of the entire organization

The 3 Levels of Data Models

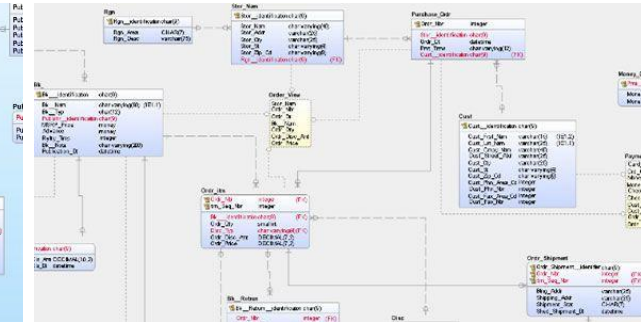
Conceptual data model



Logical data model



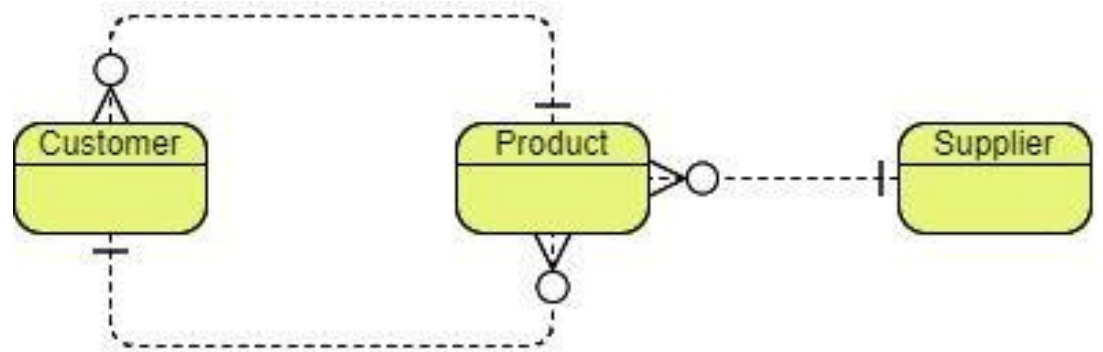
Physical data model



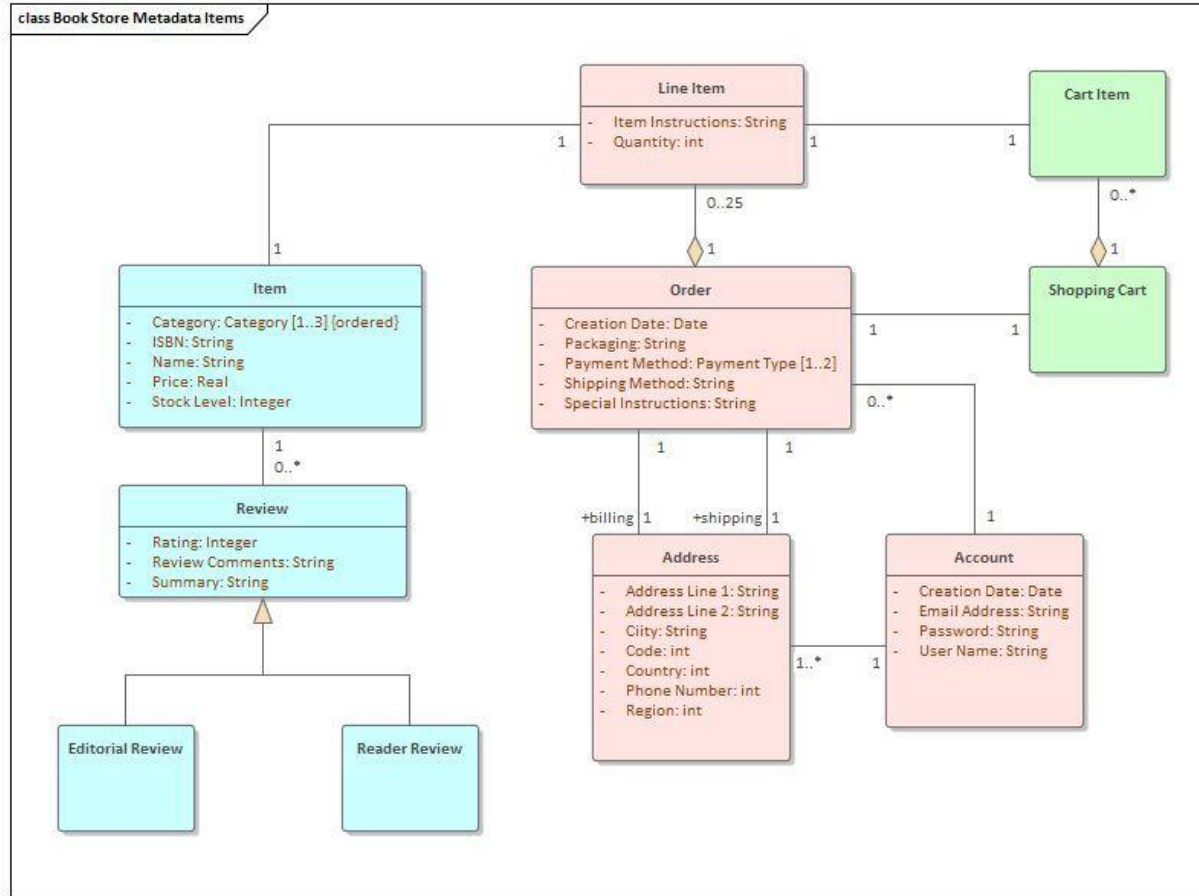
Conceptual data model

Elements of a Conceptual data model:

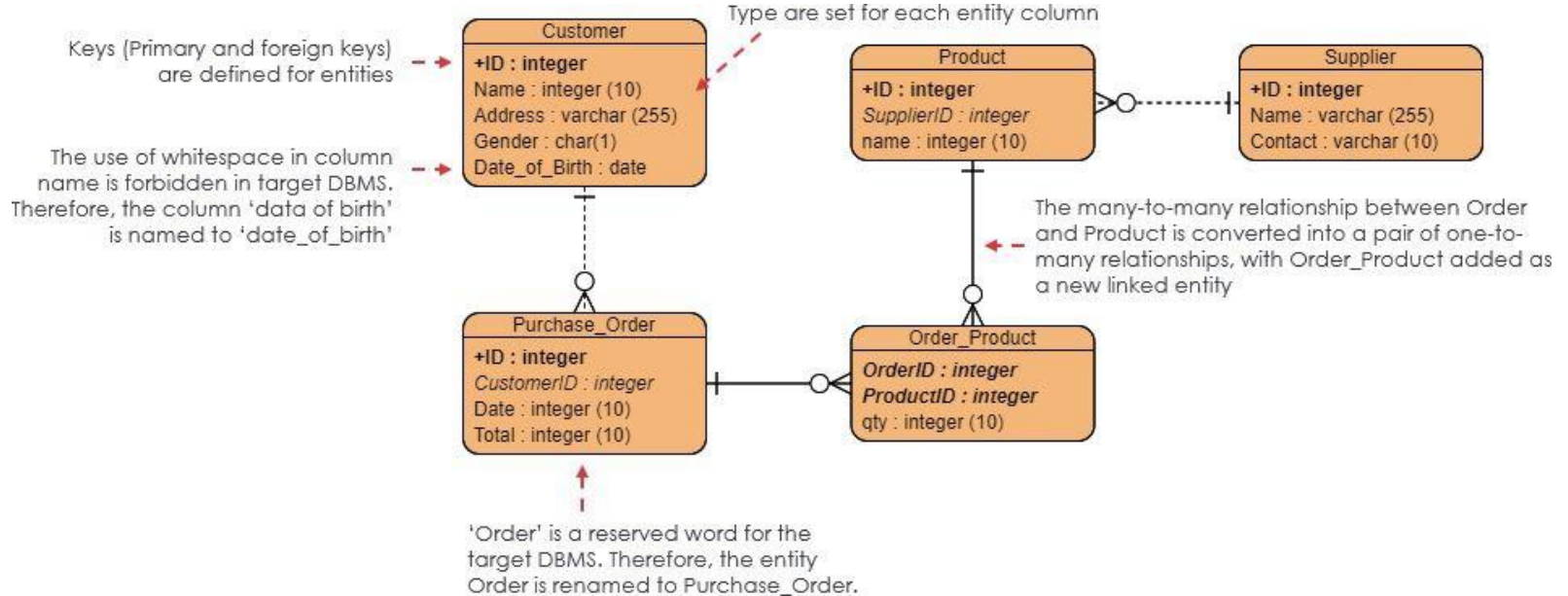
- Entity
- Attribute
- Relationship



Logical data model

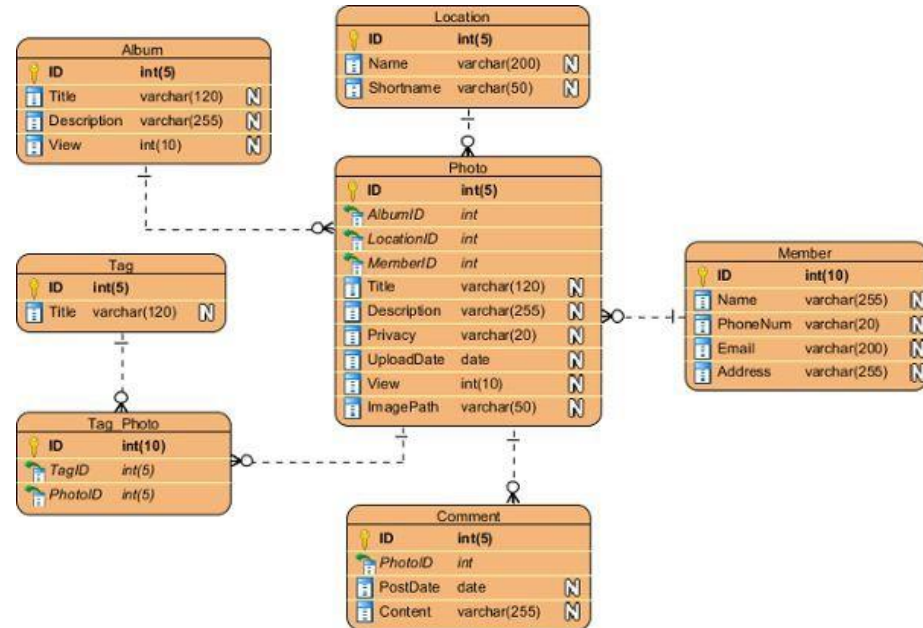


Physical data model



Data Modeling Process

1. Identify the entities
2. Identify the attributes of each entity
3. Identify relationships among entities
4. Map attributes to entities completely
5. Assign keys, decide on degree of normalization
6. Finalize and validate the data model



Benefits of Data Modeling

- Reduce errors in software and database development
- Reduced cost
- Better documentation
- Improve database performance
- Improved communication between developers and BI teams
- Ease and speed the process of database design



Data Modeling Tools

- erwin Data Modeler
- ER/Studio
- DbSchema
- ERBuilder
- HeidiSQL
- Navicat Data Modeler
- Toad Data Modeler
- SQL Database Modeler



Module 6 - Data Storage and Operations

What is Data Storage and Operations

“Data storage and Operations includes the design, implementation, and support of stored data, to maximize its value throughout its lifecycle, from creation/acquisition to disposal”

Two sub-activities:

1. Database support
2. Database technology support

Benefits of good Data Storage Strategy

1. *Reduce capital expenses*
2. *Reduce operational expenses*
3. *Easier data management*
4. *Optimized resource utilization*
5. *Easier scalability*
6. *Better performance*
7. *Better user experience*



Data Storage activities



Data Storage Management key attributes

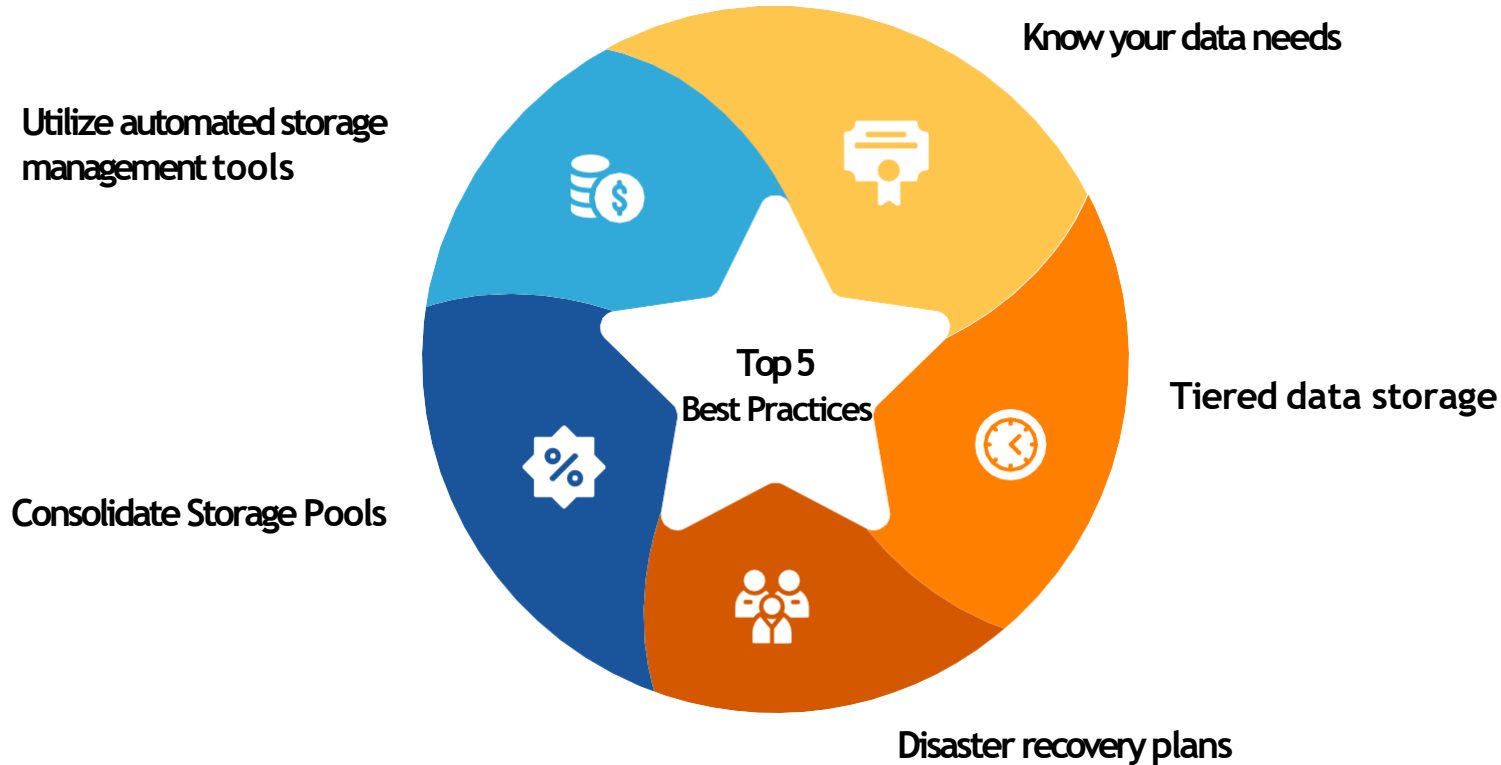
Performance

Reliability

Recoverability

Capacity

Data Storage Best Practices



Module 7 - Data Security

What is Data Security

‘**Data Security** is the process of protecting digital information from unauthorized access, corruption, or theft throughout its entire lifecycle.’

Why is Data Security important

- The average total cost of a ransomware breach is \$4.62 million, slightly higher than the average data breach of \$4.24 million ([IBM](#))
- The average per record (per capita) cost of a data breach increased by 10.3 percent from 2020 to 2021 ([IBM](#))
- The average cost of a breach with a lifecycle over 200 days is \$4.87 million ([IBM](#))
- 39 percent of costs are incurred more than a year after a data breach ([IBM](#))
- In 2021, the United States was the country with the highest average total cost of a data breach was at \$9.05 million ([IBM](#))
- 34 percent of data breaches in 2018 involved internal actors ([Verizon](#))
- It took an average of 287 days to identify a data breach ([IBM](#)).

The Goals of Data Security

1

Enabling appropriate access and preventing unauthorized access

2

Enabling compliance with regulations

3

Creating appropriate policies for privacy, protection and confidentiality

4

Ensuring that stakeholders requirements for privacy and confidentiality are met

The Principles of Data Security

1 Collaboration

2 Enterprise approach

3 Proactive Management

4 Clear accountability

5 Metadata-driven

6 Reduce risks by reducing exposure

Types of Data Security



Data Security Risks

- Accidental Data Exposure
- Phishing
- Malware
- Insider Threats
- Password Attack
- Denial-of-Service (DOS)
- Man-in-the-Middle (MITM)
- SQL Injections
- Zero-day Exploit



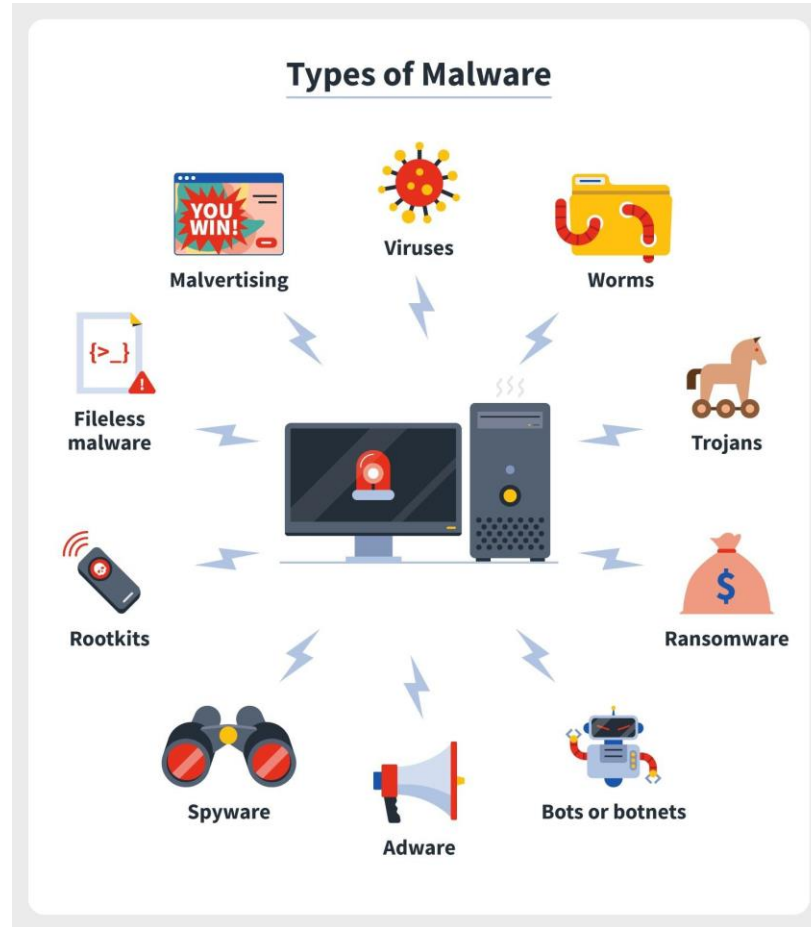
Accidental Data Exposure



Phishing

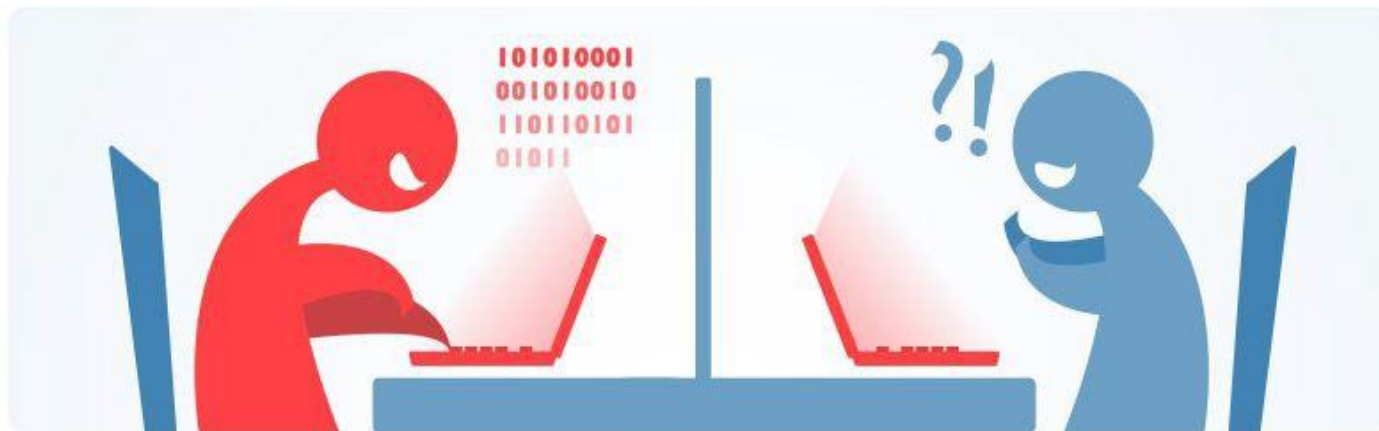


Malware



Insider Threats

Insider threat classification by CA Technologies



Malicious insiders

Intentionally use their access to sensitive data to harm the company

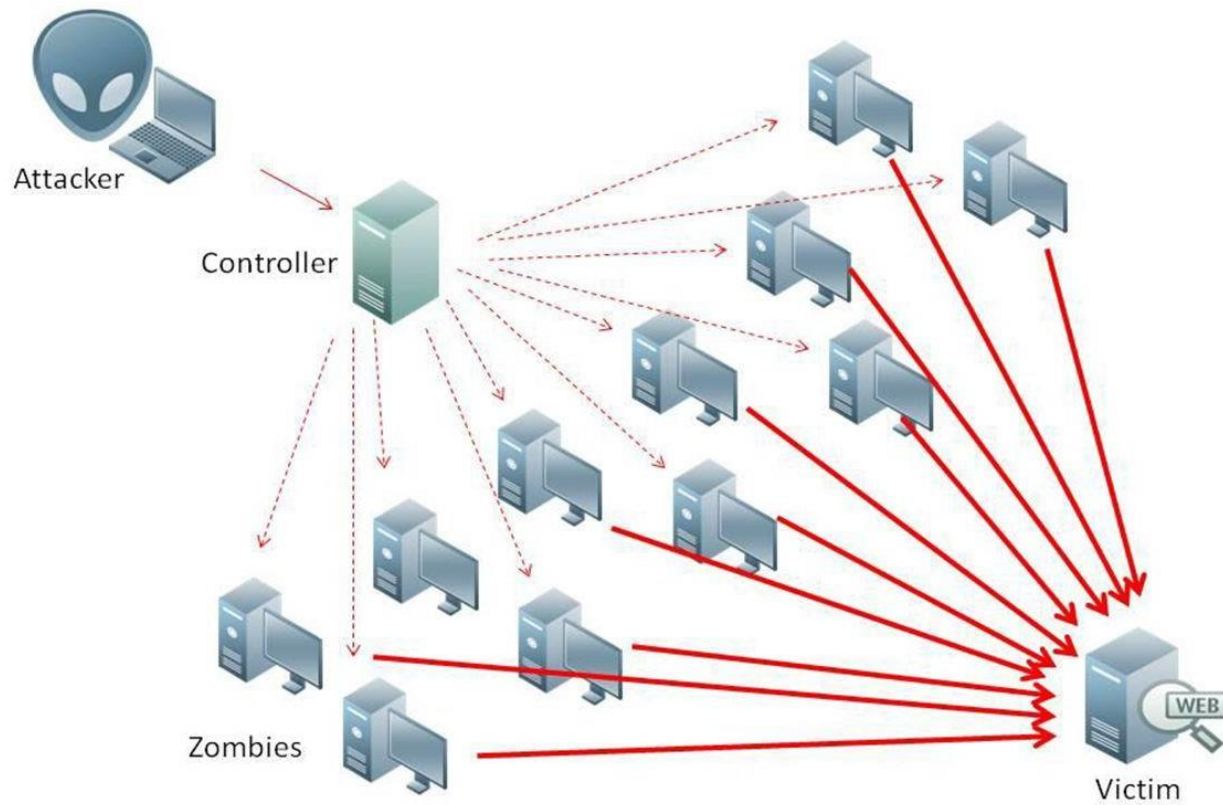
Inadvertent insiders

Cause damage to the company unintentionally

Password Attacks



Denial-of-Service (DOS)

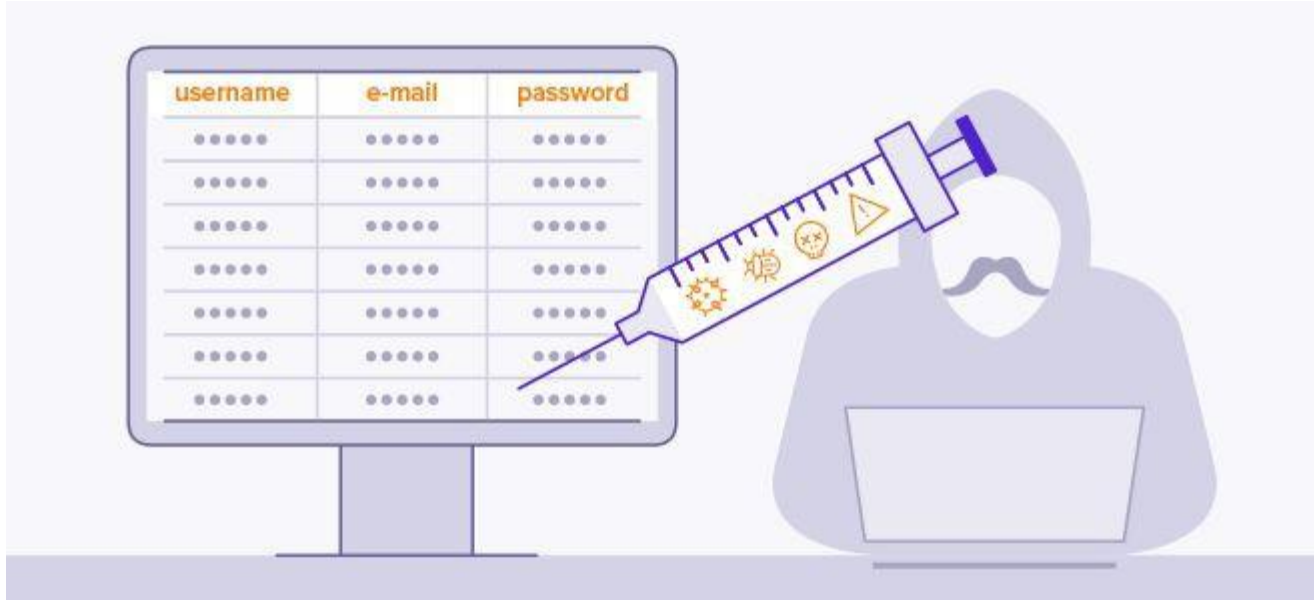


Man-In-The-Middle

HOW MAN IN THE MIDDLE ATTACKS WORK

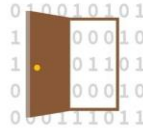


SQL Injections



Zero-Day Exploit

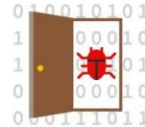
'Zero-Day' Defined



A **zero-day vulnerability** is a security software flaw that's unknown to someone interested in mitigating the flaw.



A **zero-day attack** is when hackers leverage their zero-day exploit to commit a cyberattack.



A **zero-day exploit** is when hackers take advantage of a zero-day vulnerability for malicious reasons.

Data Security Activities

1

Identify Data Security Requirements

2

Define Data Security Policy

3

Define Data Security Standards

Data Security Activities

1

Identify Data Security
Requirements

1.1

Business Requirements

1.2

Regulatory Requirements

Data Security Activities

2.1 Enterprise Security Policy

2 Define Data Security Policy

2.2 IT Security Policy

2.3 Data Security Policy

Data Security Activities

3

Define Data Security Standards

3.1

Define Data Confidentiality Levels

3.2

Define Data Regulatory Categories

3.3

Define Security Roles

3.4

Assess Current Security Risks

3.5

Implement Controls and Procedures

Data Security Tools



9 Best Practices to secure your data

1. Employees education
2. Create Insider Threat Policies
3. Phishing Simulations
4. Backup data
5. Update Systems and Software
6. Utilize HTTPS
7. Maintain Compliance
8. Use multi-factor authentication
9. Employe latest secure coding practices



Module 8 - Data Integration & Interoperability

What is Data Integration

‘**Data Integration** is the process of consolidating data from different sources into one, unified view for efficient data management”

Example of Data Integration

Company A uses the following:

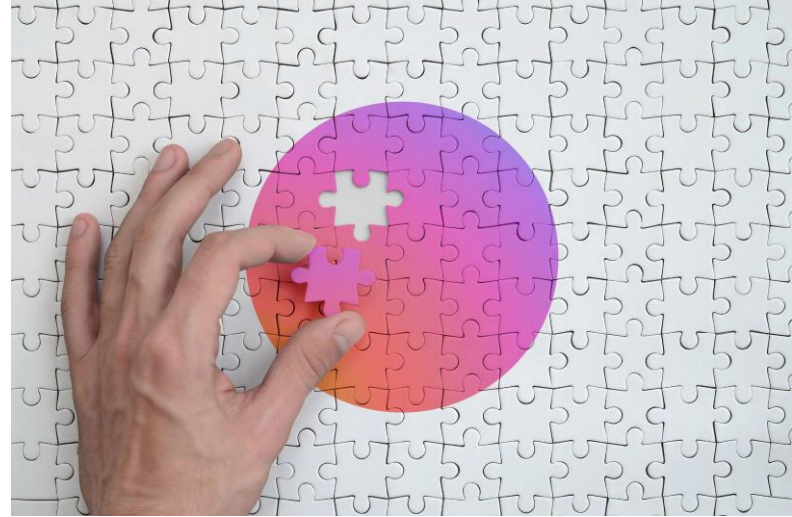
- Salesforce for customer information and sales pipeline data
- External vendor for additional customer firmographics
- Internal database that tracks customer satisfaction ratings from surveys
- Internal financial system that tracks the sales revenue per customer
- Marketing department database on customer campaigns

Goal: Integration all the above sources into single source!



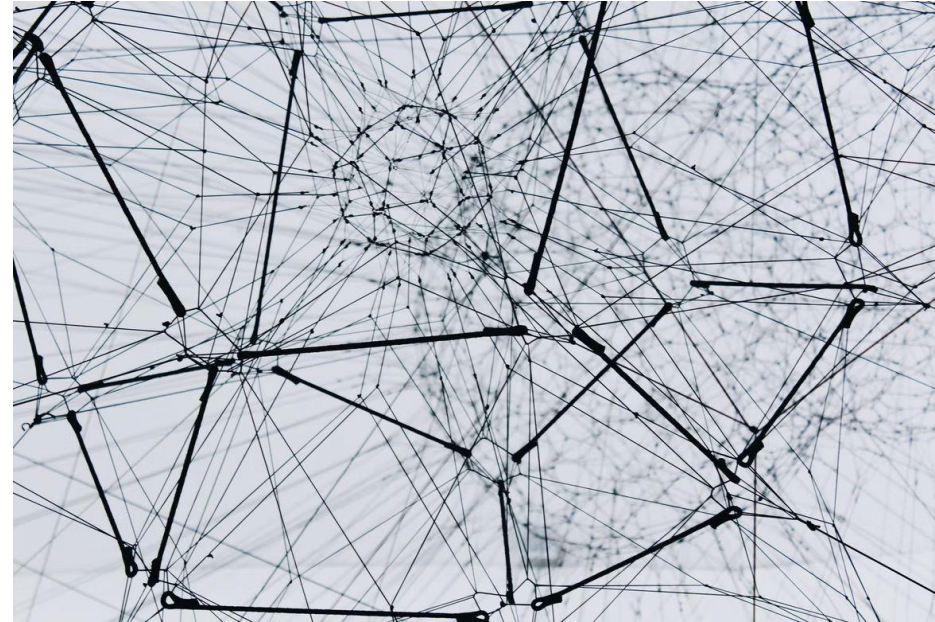
Importance of Data Integration

- Complete view of business intelligence, insights and analytics
- Increased efficiency and ROI - no need for manual data gathering
- Better employee, customer and partner experience
- Improves Collaboration
- Eliminates Data Silos
- Reduces errors
- Faster innovation, sales, time to market
- Improves Data Quality and Integrity



Techniques for Data Integration

- Manual Data Integration
- Middleware Data Integration
- Application Based Integration
- Uniform Access Integration
- Common Storage Integration (Data Warehousing)
- Data Virtualization



Manual Data Integration

Pros:

- Low cost
- Total control

Cons:

- Difficult to scale
- Human error



Middleware Data Integration

Pros:

- Better data streaming
- Easier access between systems

Cons:

- Maintenance
- Limited functionality



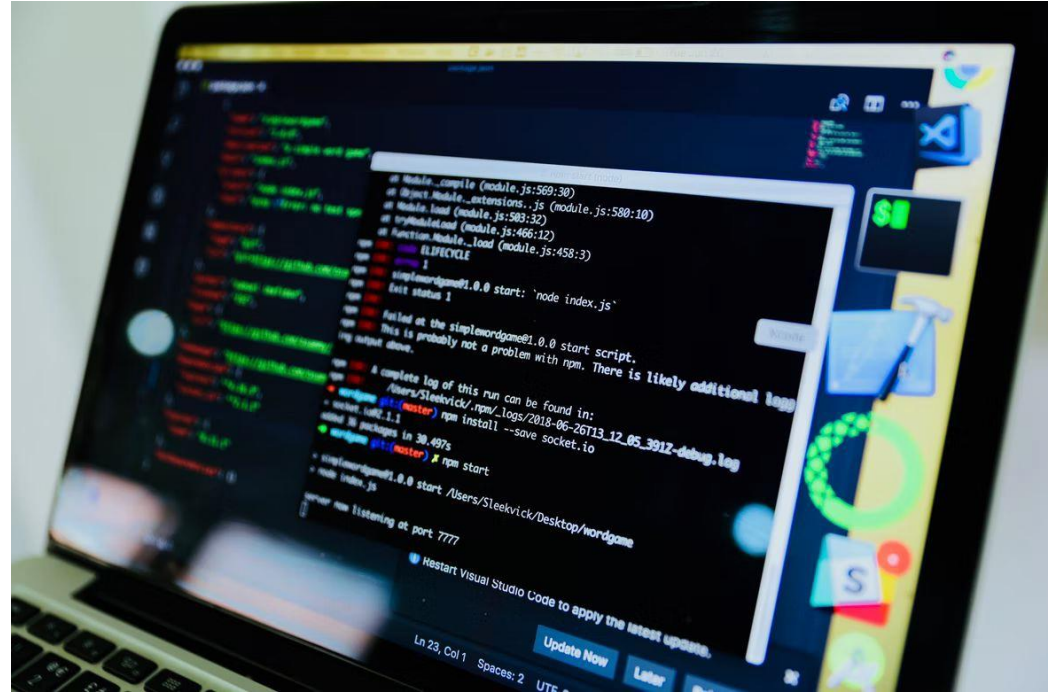
Application Based Integration

Pros:

- Simplified processes
- Easier information exchange
- Fewer resources used

Cons:

- Maintenance
- Inconsistent results
- Complicated setup
- Difficult data management



Uniform access Integration

Pros:

- Lower storage requirements
- Easier data access
- Simplified view of data

Cons:

- Data integrity challenges
- Strained systems



Common Storage Integration (Data Warehousing)

Pros:

- Reduced burden
- Increased data version management control
- Cleaner data
- Enhanced data analytics

Cons:

- Increased storage costs
- Higher maintenance costs



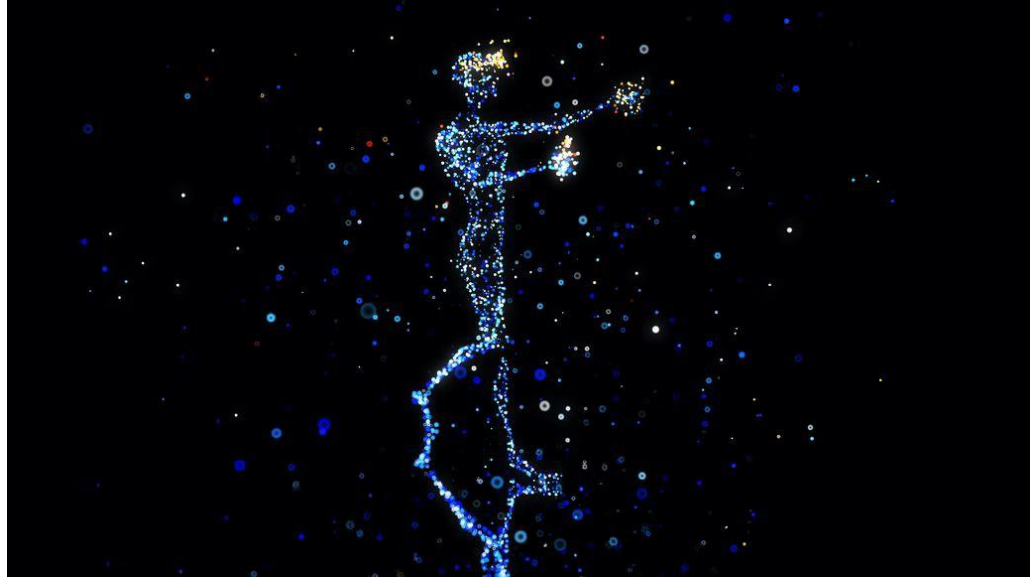
Data Virtualization

Pros:

- No need to move data
- Scalable
- No need to maintain data in multiple locations

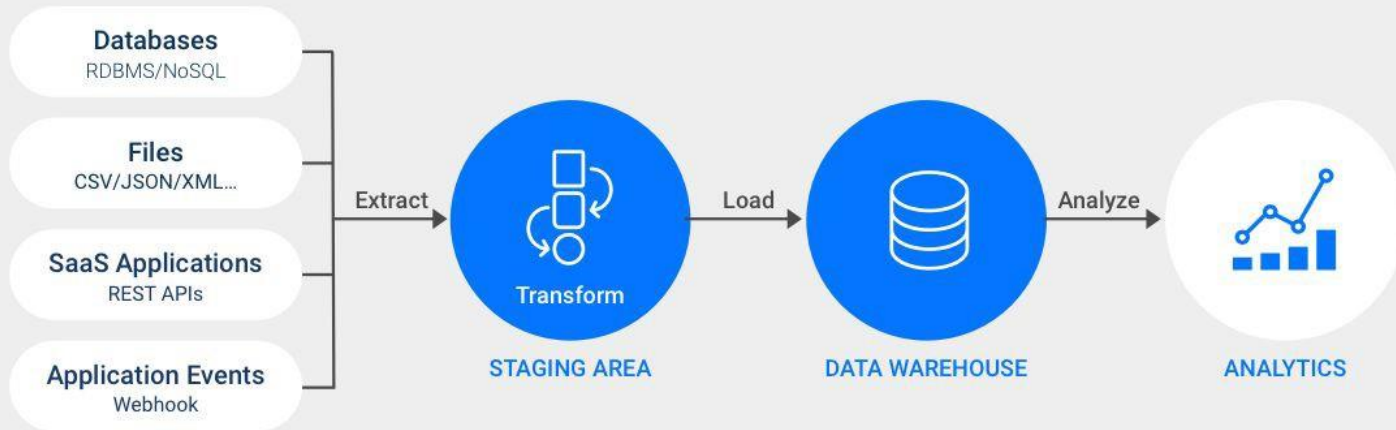
Cons:

- Cost



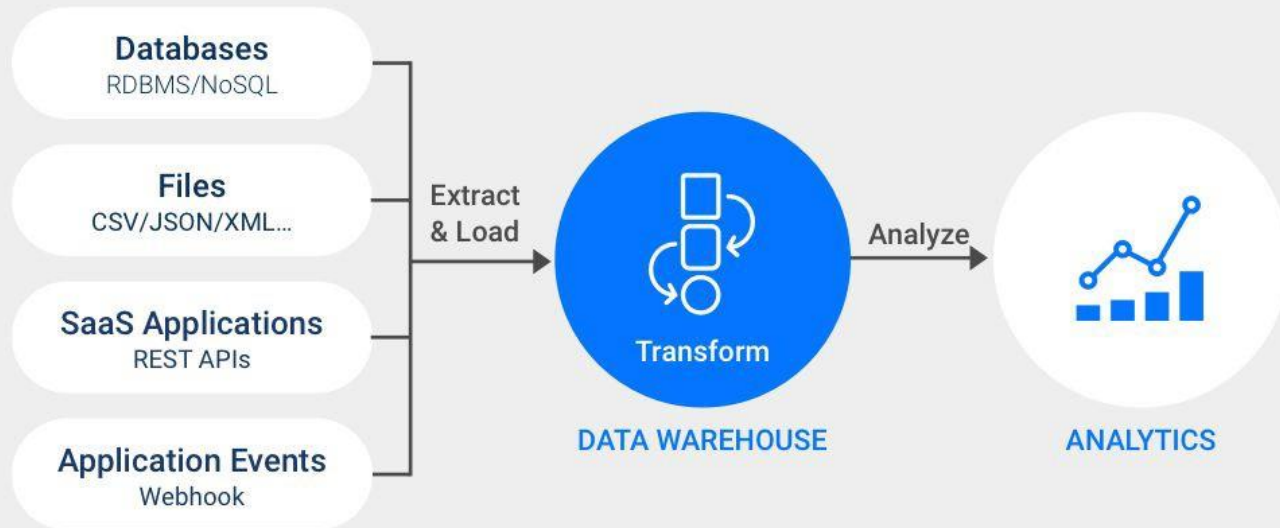
ETL

ETL PROCESS



ELT

ELT PROCESS



ETL vs ELT

Pros of ETL:

- **Compliance** - ETL is better for compliance with GDPR, HIPAA, CCPA and other standards
- **Implementation** - many ETL tools and experts
- **Maturity** - more documentation, tools and best practices

Pros of ELT:

- **Maintenance** - all data is always flowing with automated process
- **Speed** - data available in Data Warehouse faster since no transformation layer
- **Cost** - only cloud-based platforms needed at lower cost. On premise ETL processes require expensive hardware



Data Integration Tools

On-premise tools

- Oracle Data Service Integrator
- Informatica PowerCenter
- IBM InfoSphere Information Server

Cloud-based tools

- SnapLogic
- Talend Cloud Integration

Open-source tools

- Talend Open Studio
- Tibco Jaspersoft



Data Integration Best Practices

1. Identify your business needs first
2. Include internal business expert in the data integration team
3. Consider Long-Term Goals
4. Take into consideration the total cost of all methods
5. Avoid very complex data integration solutions
6. Choose a flexible solution



Module 9 - Document & Content Management

What is Document & Content Management

‘**Document & Content management** is the process of establishing planning, implementation and control activities for lifecycle management of data and information found in any form or medium - outside of relational databases’

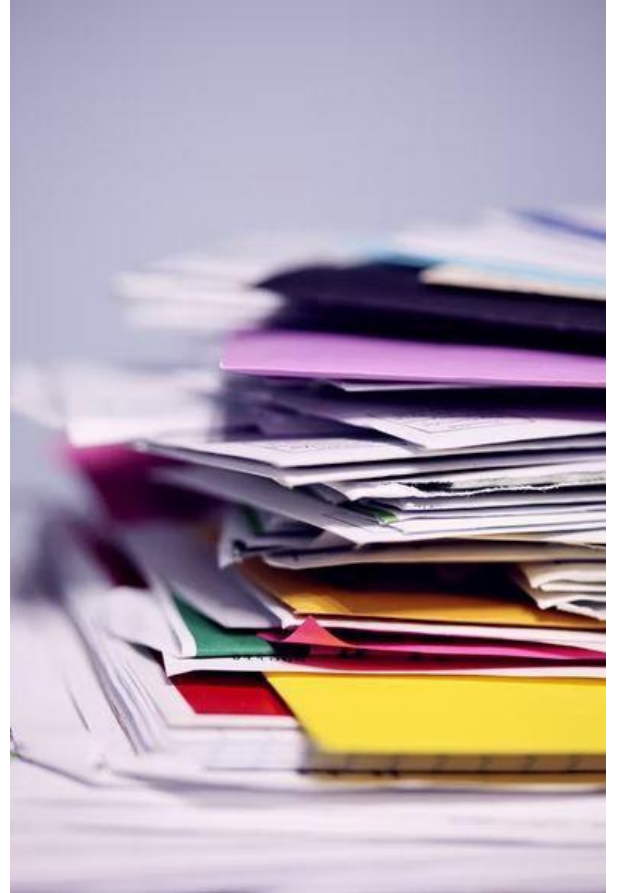
Why do we need Document & Content Management?

- Comply with legal obligations
- Comply with customer expectations regarding records management
- Effective and efficient storage, retrieval and use of documents and content
- Integration between structured and unstructured content



What is a DMS?

- DMS stands for Document Management System
- What are the benefits?
- Types of DMS



What is a CMS?

- CMS stands for Content Management System
- What are the benefits?
- Types of CMS



ECMS

- ECMS stands for Enterprise Content Management System

- What are the benefits?

- CMS vs ECM



Figure 1: Magic Quadrant for Content Services Platforms



Source: Gartner (October 2021)

Document Management vs Enterprise Content Management

Comparison	Document Management System (DMS)	Enterprise Content Management System (ECMS)
Type of Data	Structured data in traditional formats (Word, PDF, PowerPoint, Excel, etc)	Structured + unstructured data such as images, audio, video files, HTML, etc
Main purpose	Workflow management and regulatory compliance	Storage, retrieval and publishing of content
Key difference	DMS is a software	ECM is a set of tools and processes. ECM is a broader version of DMS
Company size	DMS only solution can work well for small companies	ECM solution needed in bigger organizations

Module 10 - Master & Reference Data Management

What is Master Data

DAMA Guide to Data Management Body of Knowledge: “Master Data represents data about the business entities that provide context for business transactions”

Gartner: ‘Master Data is the consistent and uniform set of identifiers and extended attributes that describes the core entities of the enterprise including customers, prospects, citizens, suppliers, sites, hierarchies and chart of accounts”

What is Reference Data

DAMA Guide to Data Management Body of Knowledge: “Reference data is data used to classify or categorize other data”

Examples of Reference Data:

- Postal codes
- Language codes
- Customer segments
- Country codes
- Cost centers

Master Data vs Reference Data

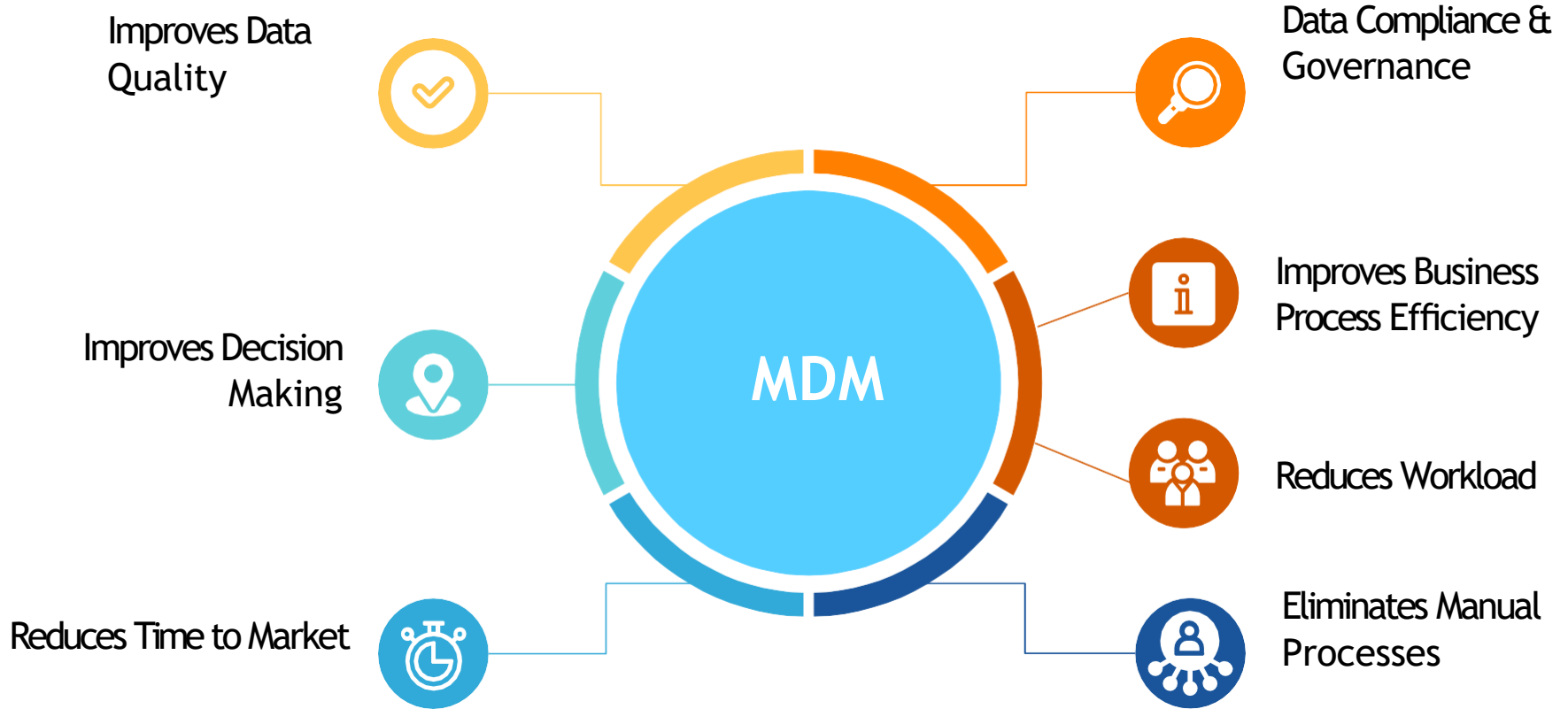
Comparison	Master Data	Reference Data
Main purpose	Represents the business objects which contain the most valuable, agreed upon information shared across the organization	Data that defines the set of permissible values to be used by other data fields
More on usage	Master data is the data shared by multiple systems, applications, processes in the organization	Reference data is a type of master data that is used by other data fields
Examples	• Customer information - names, phone numbers and addresses • Product information - product name and location • Partner data - partner name and address	• Fixed conversion rates - weight, temperature, length, etc • Currency codes • Language codes • Customer Segments • Cost centers • Postal codes • Units of measurement

What is Master Data Management (MDM)

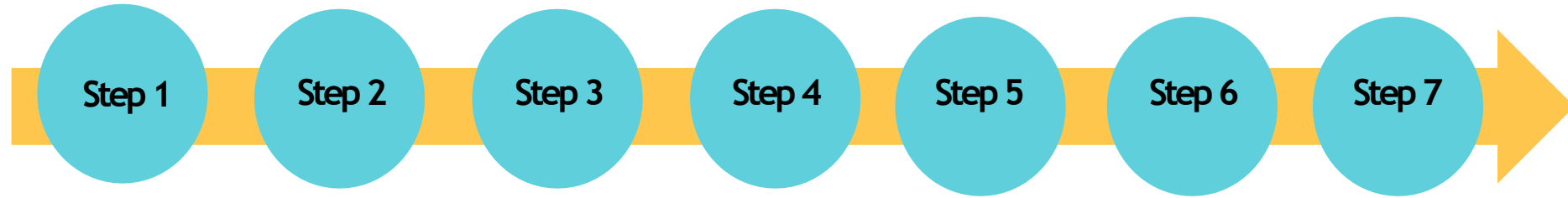
Master Data Management is the process of creating and maintaining a single master record - or single source of truth - for each person, place, and thing in a business.

Through MDM, organizations gain a trusted, current view of key data that can be shared across the business and used for better reporting, decision-making, and process efficiency.

Why MDM is important



Steps to implement MDM



Learn about MDM:

- Important Capabilities
- Implementation Styles
- Domain Categories

Analyze your current
data environment

Define your business
goals

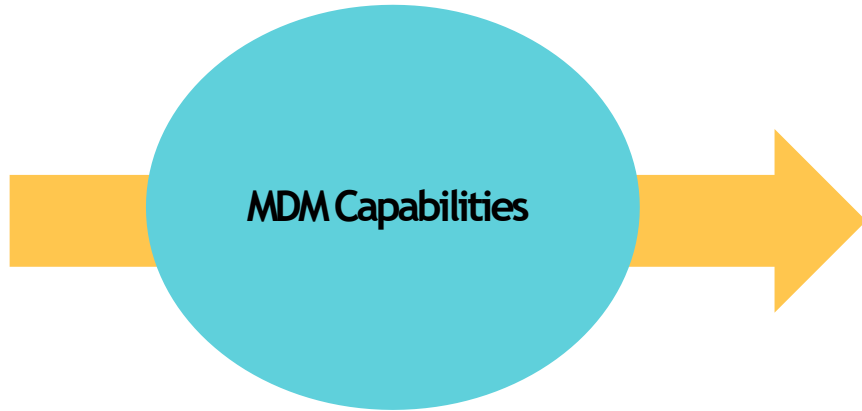
Secure the right
sponsorship

Get the right team and
tool

Implement a
quick win

Repeat Step 6 and
monitor

MDM Solution Capabilities



- Workflow/BPM
- Loading/Sync/Business Services
- Data Modeling
- Information Quality/Semantics
- Perform/Scale/Availability/Security
- Hierarchy Management
- Data Stewardship
- Data Governance
- Multiple Implementation Styles
- Multiple Usage Scenarios
- Multiple Domain and Multidomain
- Product Suite Internal Integration

MDM Implementation Styles

Consolidation



- **Ideal for reporting or analytics that reside in a BI/data warehouse**
- Nonintrusive to the business
- BI is the business platform
- Any industry
- Benefits dependent on success of BI strategy
- No attempt to clean up source data

Registry



- **Low control, autonomous environments**
- Nonintrusive to edge applications
- Emphasis is on remote data and application-to-application integration (lots of real-time network access)
- Distributed governance
- Faster to implement than coexistence and centralized

Coexistence



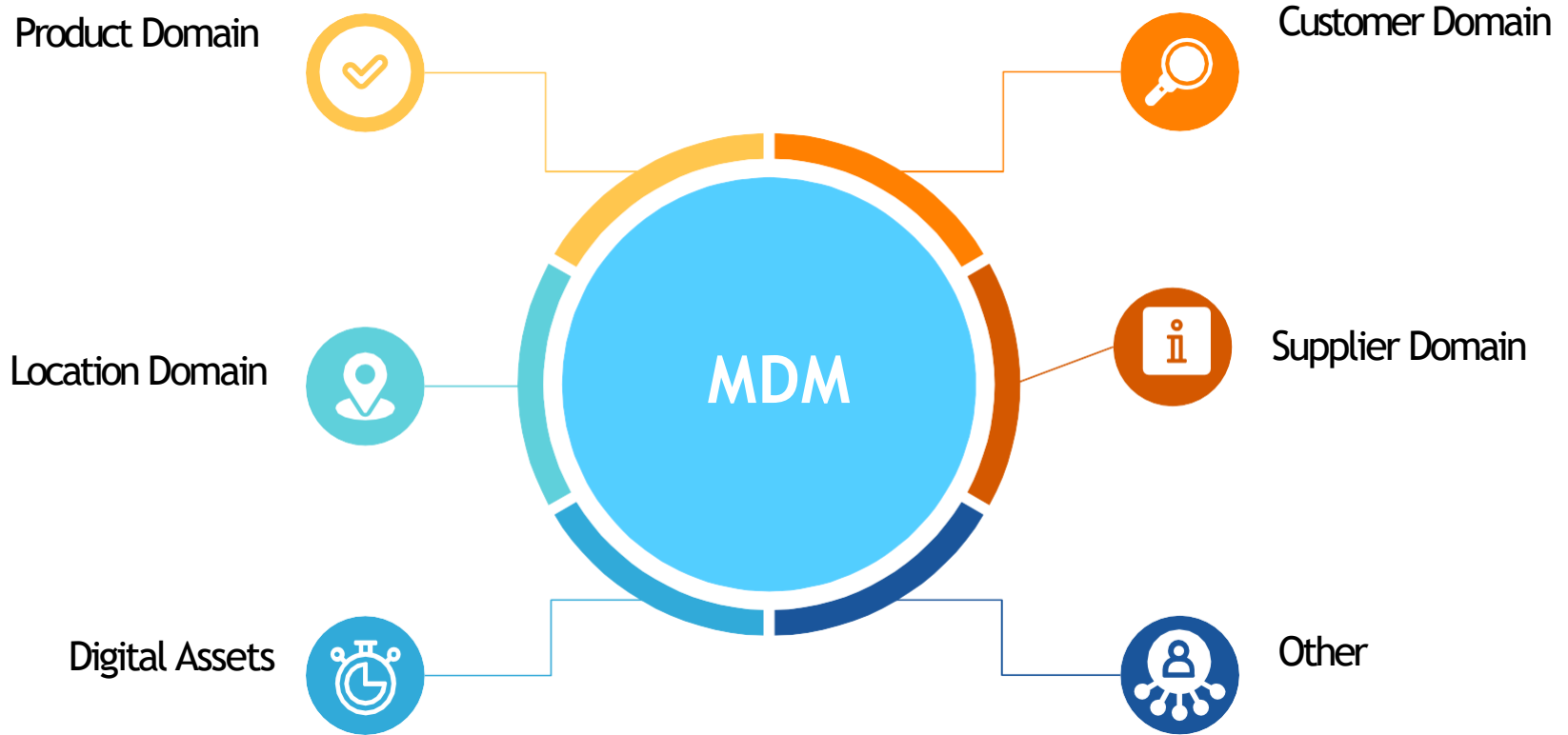
- **Large-scale distributed model**
- Largest change to information infrastructure
- Greatest need to mirror data
- Global and local governance
- Greatest risk over control, security
- Focused on shared services

Centralized

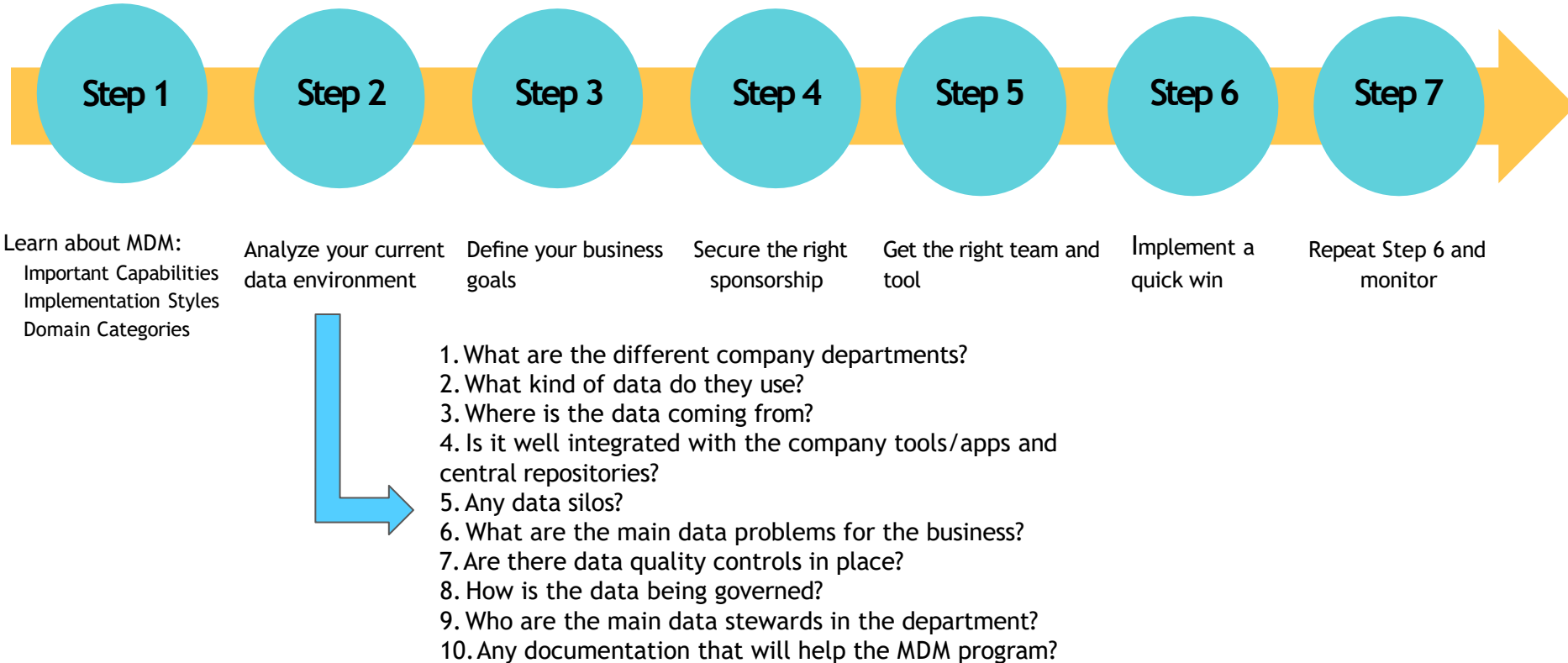


- **High-control, top-down environments**
- Largest change to application infrastructure
- Hugely invasive to the business
- Centralized governance
- Greatest control over access, security
- Focus on common services

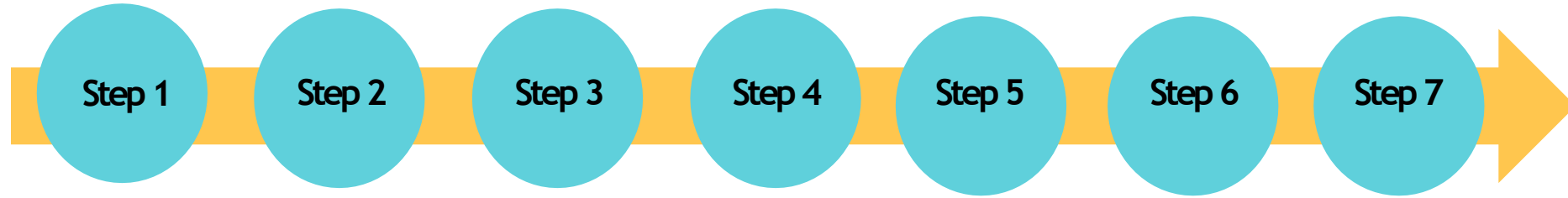
MDM Domains



Steps to implement MDM



Steps to implement MDM



Learn about MDM:
Important Capabilities
Implementation Styles
Domain Categories

Analyze your current
data environment

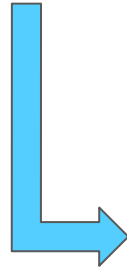
Define your business
goals

Secure the right
sponsorship

Get the right team and
tool

Implement a
quick win

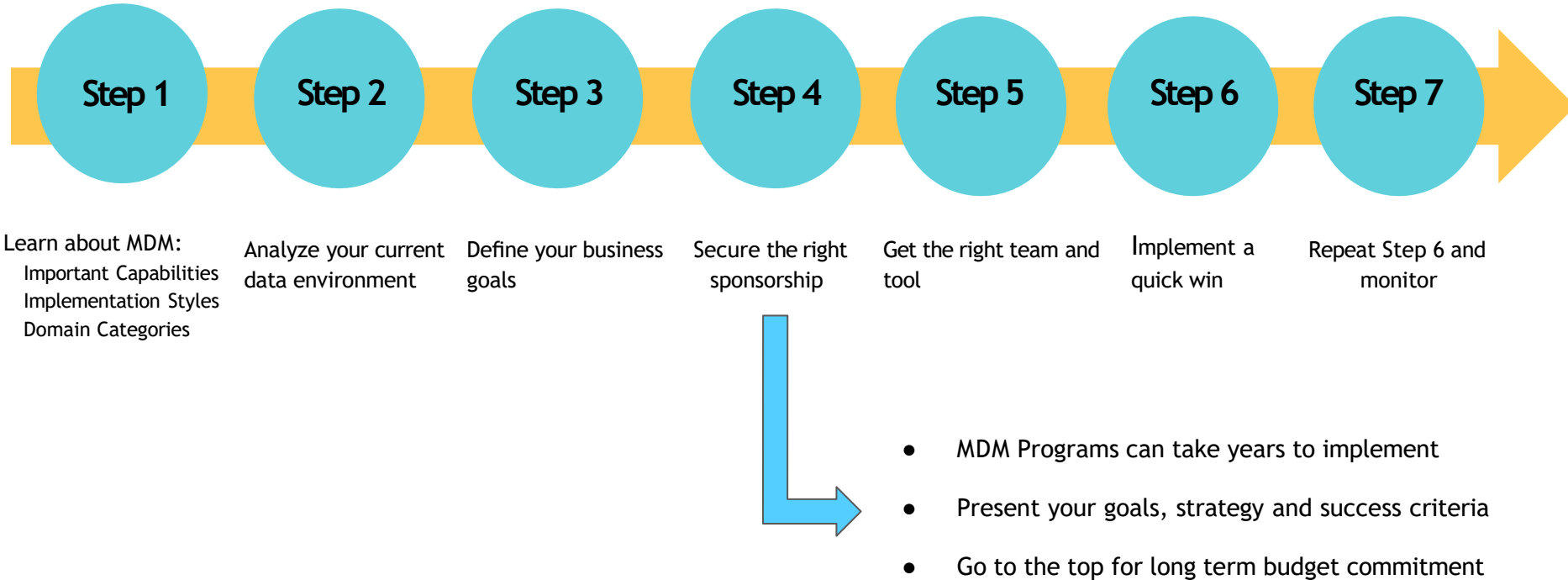
Repeat Step 6 and
monitor



Some business drivers for MDM include:

- Up sell and cross sell opportunities
- Complete view of customers
- Improved Data Quality and business decisions
- Centralization of data
- Reduce costs of data maintenance and support
- Improved customer experience
- Other

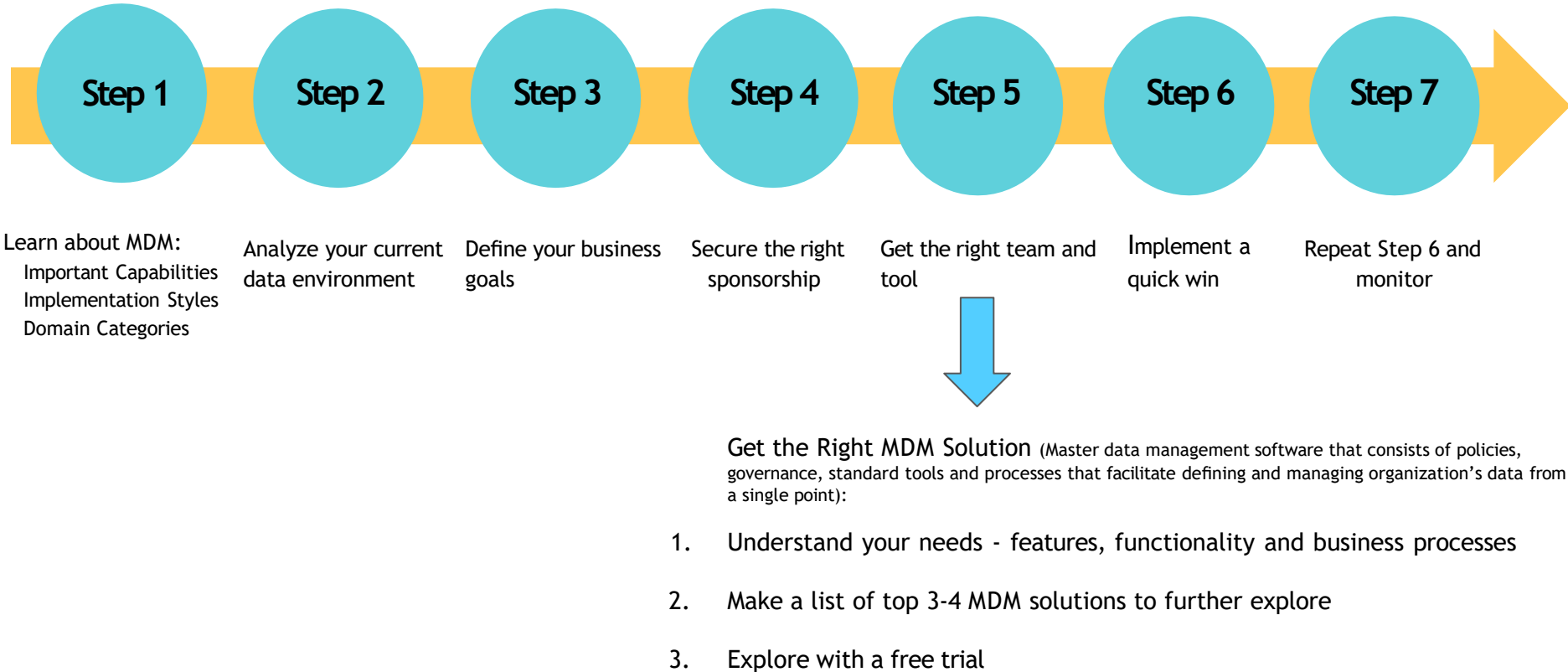
Steps to implement MDM



Steps to implement MDM



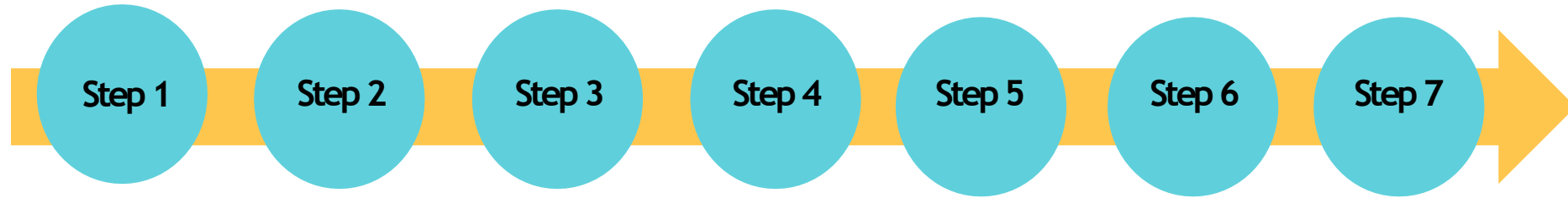
Steps to implement MDM



Steps to implement MDM



Steps to implement MDM



Learn about MDM:
Important Capabilities
Implementation Styles
Domain Categories

Analyze your current
data environment

Define your business
goals

Secure the right
sponsorship

Get the right team and
tool

Implement a
quick win

Repeat Step 6 and
monitor



- Maintain the long term vision and release “wins” at regular intervals
- Keep on tracking the KPIs
- Maintain leadership’s interest in the MDM Program

Module 11 - Data Warehousing & Business Intelligence

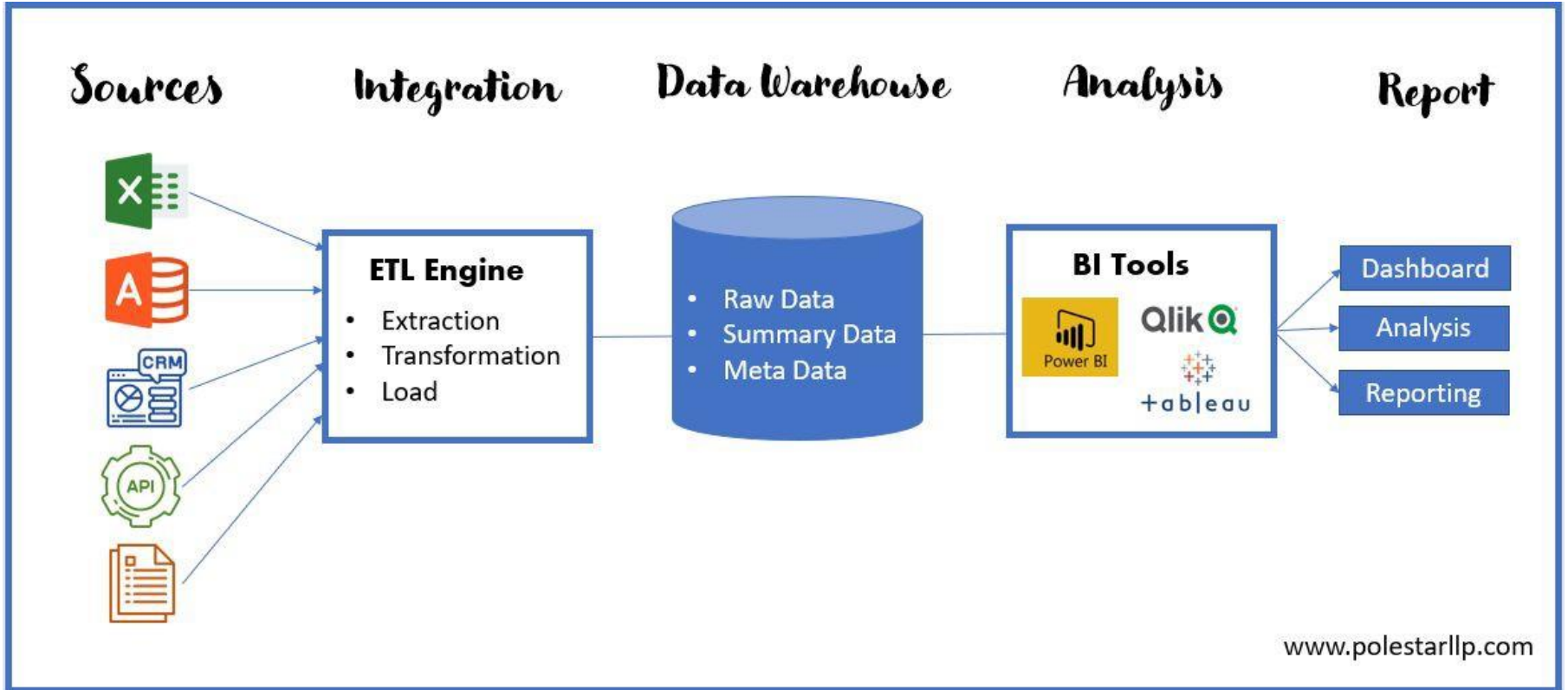
What is Data Warehousing and Business Intelligence

“Data warehousing is the secure electronic storage of information by a business or other organization. The goal of data warehousing is to create a collection of historical data that can be retrieved and analyzed to provide useful insight into the organization's operations

Data warehousing is a vital component of business intelligence. That wider term encompasses the information infrastructure that modern businesses use to track their past successes and failures and inform their decisions for the future.

”

How the process works

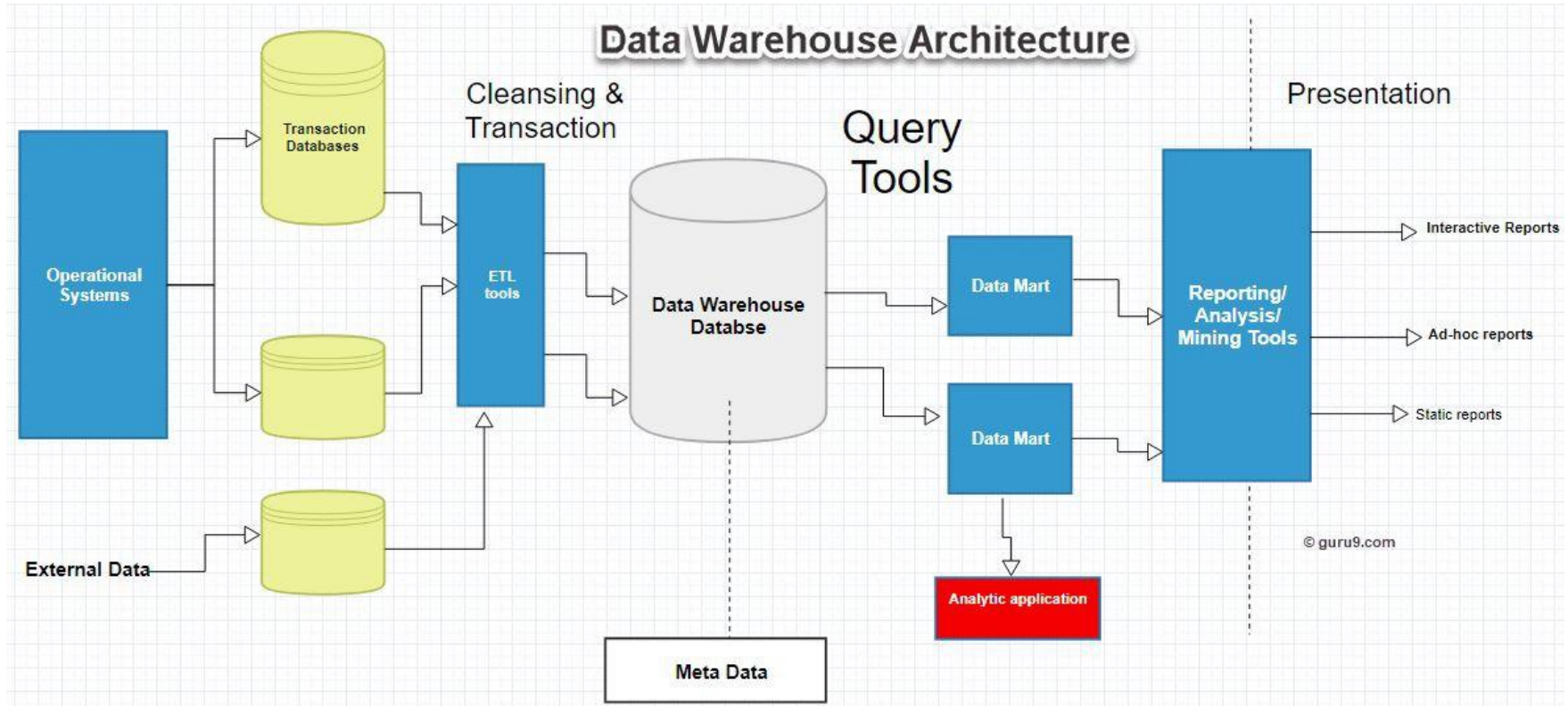


What is a Data Warehouse (DW/DWH)

Data Warehouse (DW or DWH), and sometimes also referred to as Enterprise Data Warehouse (EDW) is a central repository of integrated data from one or more different sources.

Data Warehouses are used to store current and historical data in one single place.

Data Warehouse Components



What is a Data Warehouse (DW/DWH)

Data Warehouse (DW or DWH), and sometimes also referred to as Enterprise Data Warehouse (EDW) is a central repository of integrated data from one or more different sources.

Data Warehouses are used to store current and historical data in one single place.

Data Warehouse vs Database

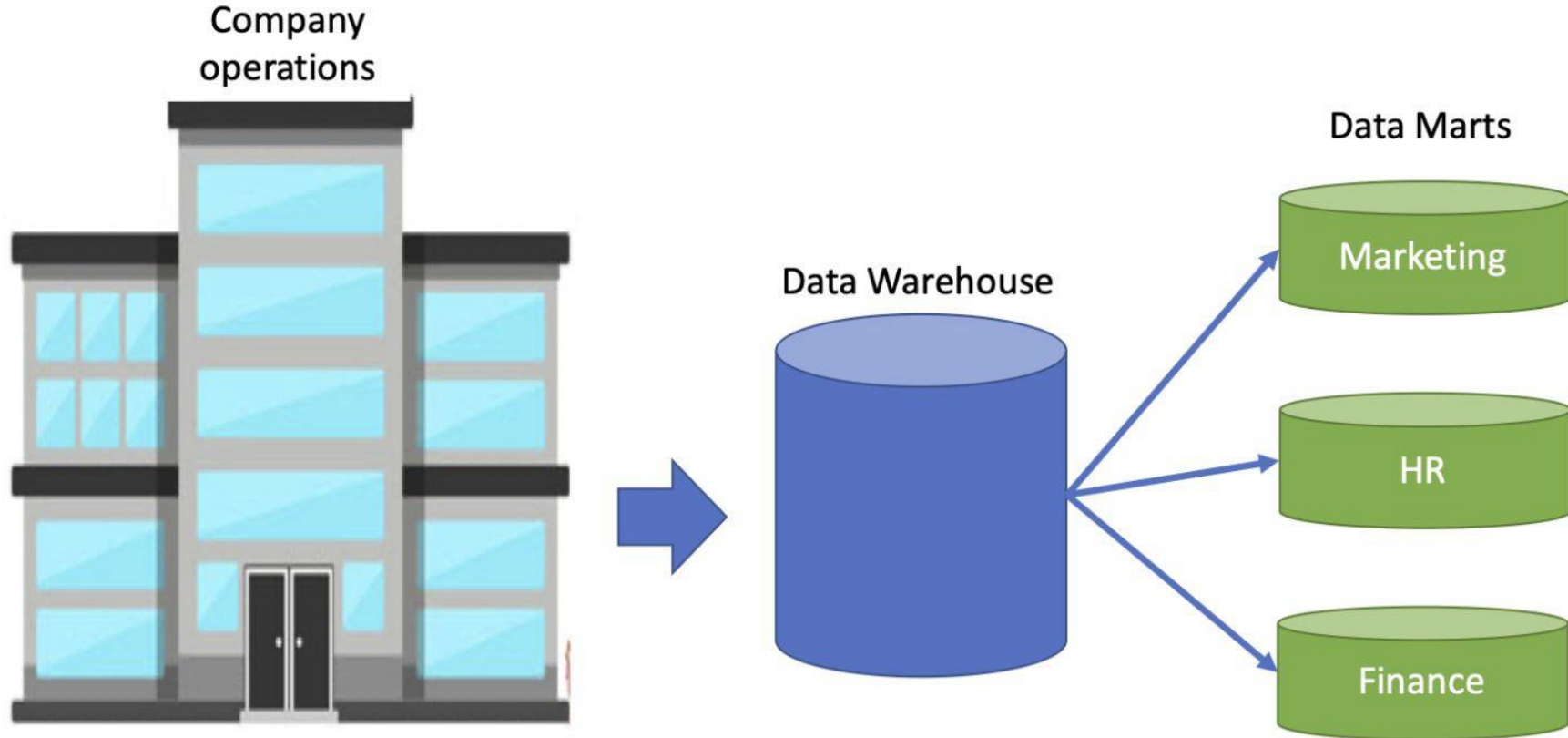
Database vs. Data Warehouse

While these two data storage elements may seem similar, they offer very different capabilities. Here is a brief breakdown of the differences:

Database	Data Warehouse
Designed to record data	Designed to analyze data
Stores detailed data	Stores summarized data
Uses Online Transactional Processing OLTP	Uses Online Analytical Processing OLAP
Performs fundamental business operations and transactions	Allows users to analyze business data
Data is available in real time	Data must be refreshed when needed
Application-oriented data collection	Subject-oriented data collection
Limited to a single application	Draws data from a range of other applications

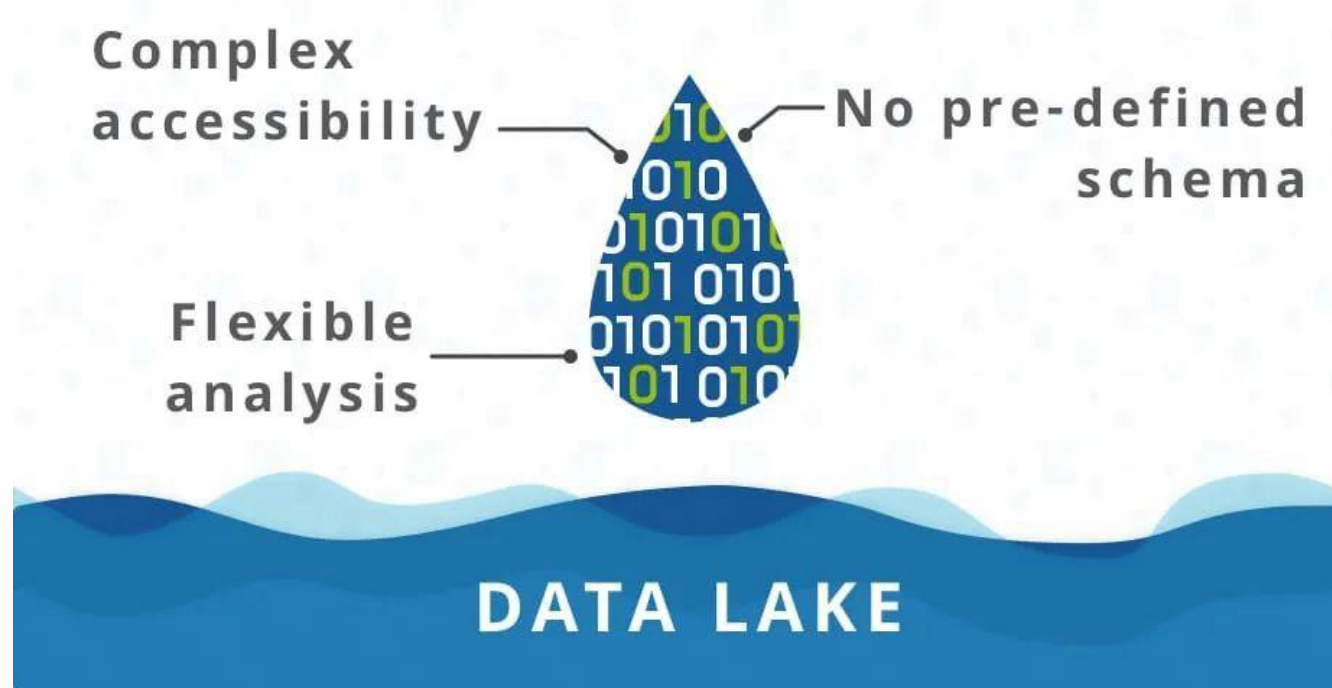
source:  SelectHub

Data Warehouse vs Data Marts

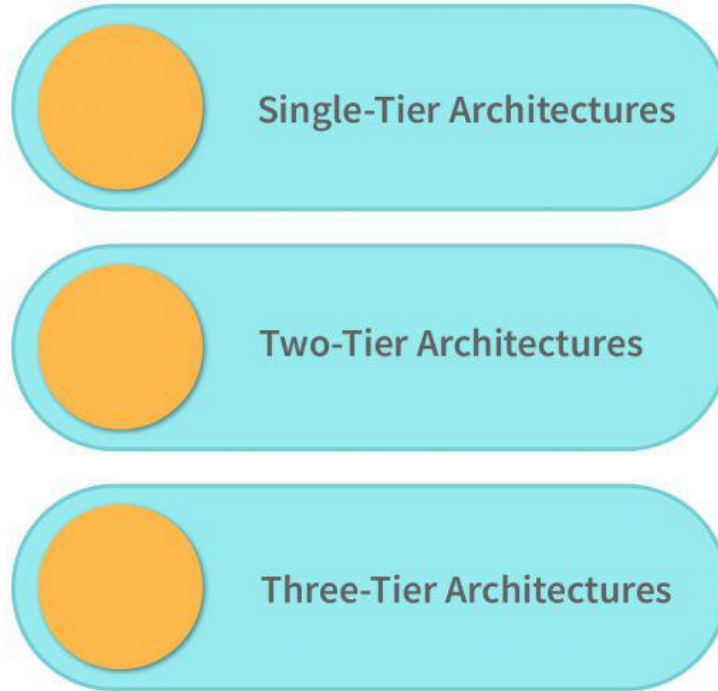


Data Warehouse vs Data Lake

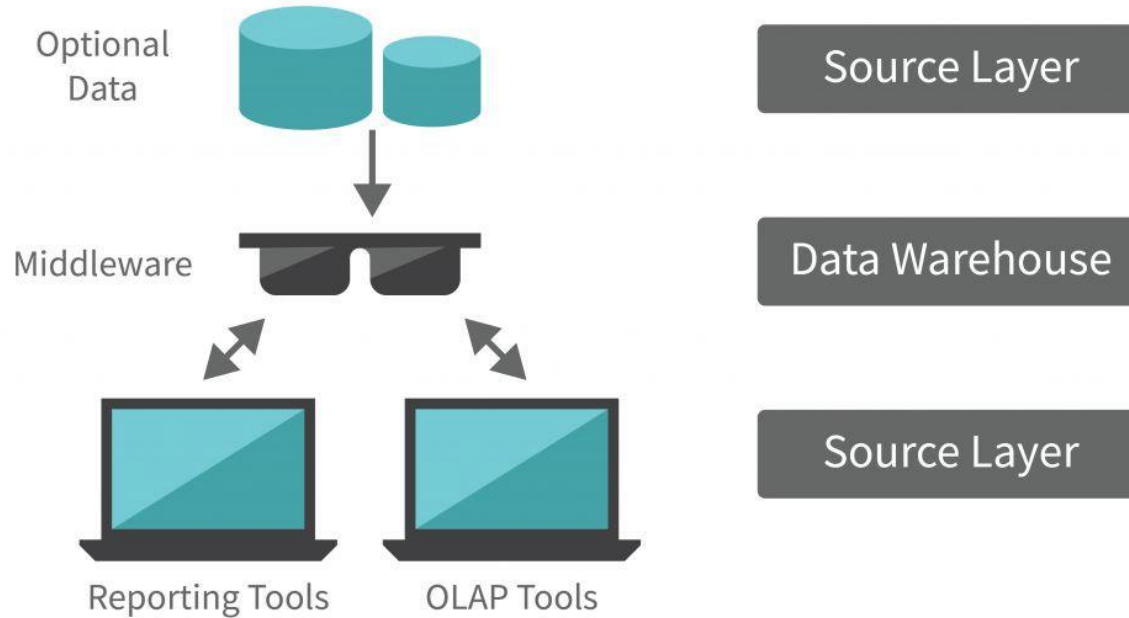
A data lake is a central storage repository that holds [big data](#) from many sources in a raw, granular format.



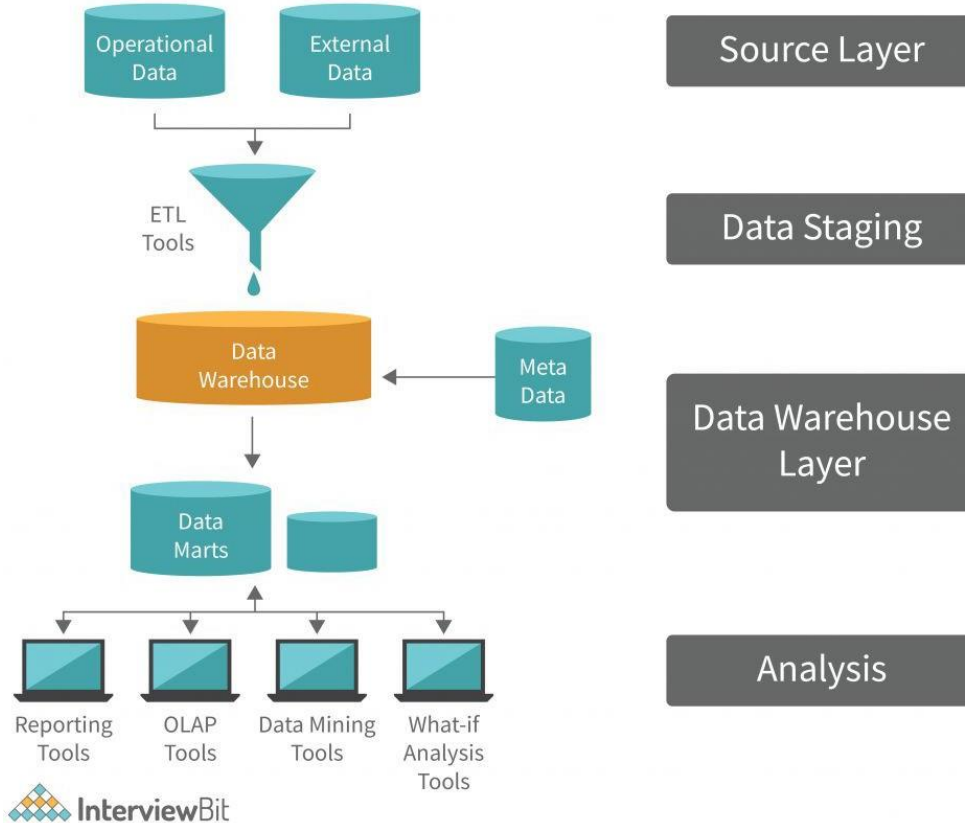
Data Warehouse Architecture Types



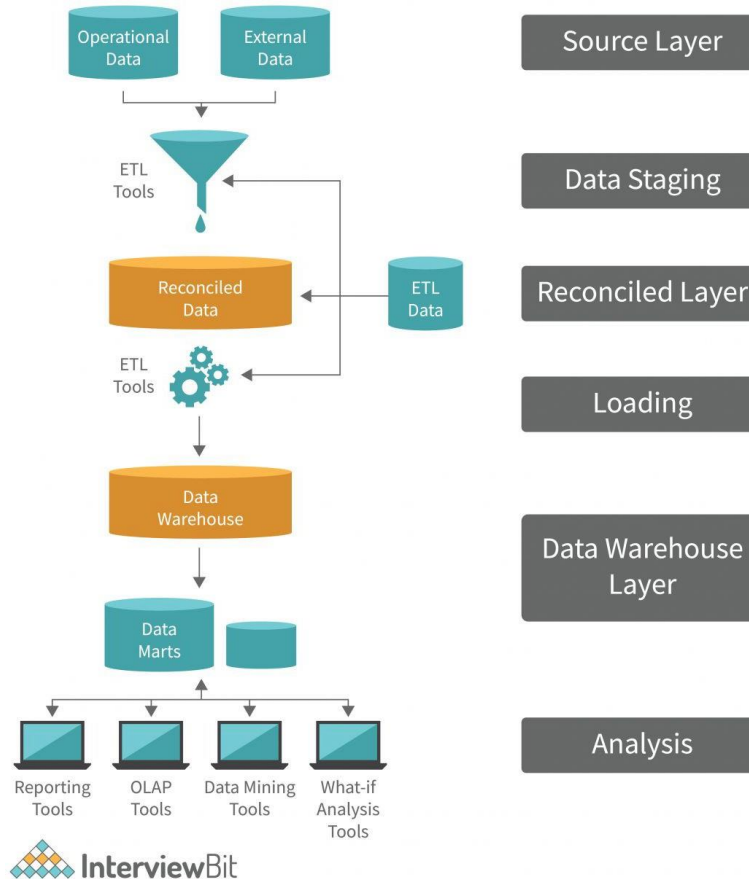
Single-Tier Data Warehouse Architecture



Two-Tier Data Warehouse Architecture



Three-Tier Architecture for a Data Warehouse System



What is Business Intelligence

“Business intelligence (BI) leverages software and services to transform data into actionable insights that inform an organization’s strategic and tactical business decisions.”

Business Intelligence vs Business Analytics

“The primary distinction between business intelligence and business analytics is the focus on when events occur. **Business intelligence is focused on current and past events that are captured in the data. Business analytics is focused on what's most likely to happen in the future..**”

Applications of Business Intelligence

1. Sales Intelligence
2. Visualization
3. Reporting
4. Performance Management



Categories of BI analysis



Descriptive

Explains what happened.



Diagnostic

Explains why it happened.



Predictive

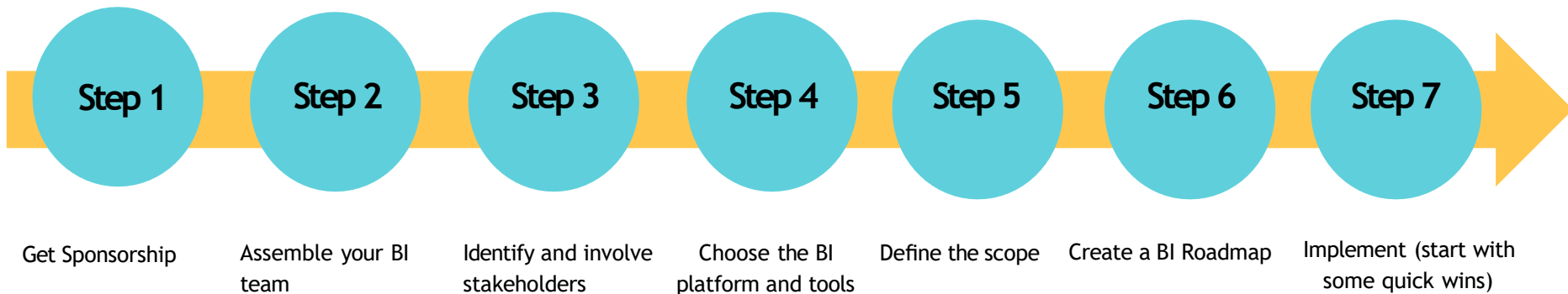
Forecasts what might happen.



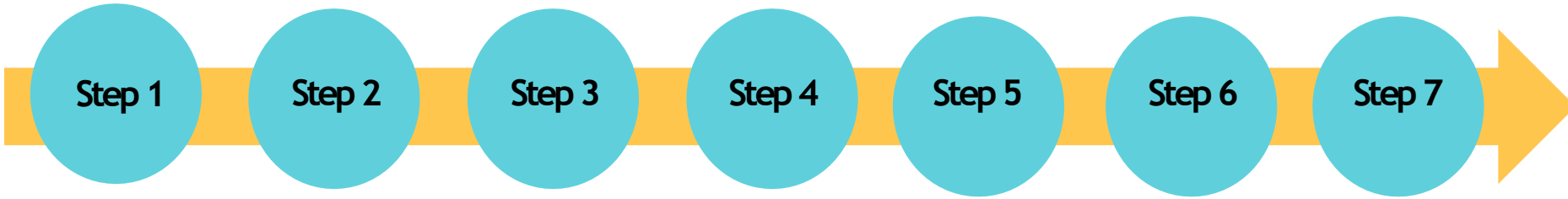
Prescriptive

Recommends an action based on the forecast.

Create a Business Intelligence Strategy



Create a Business Intelligence Strategy



Get Sponsorship

Assemble your BI team

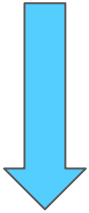
Identify and involve stakeholders

Choose the BI platform and tools

Define the scope

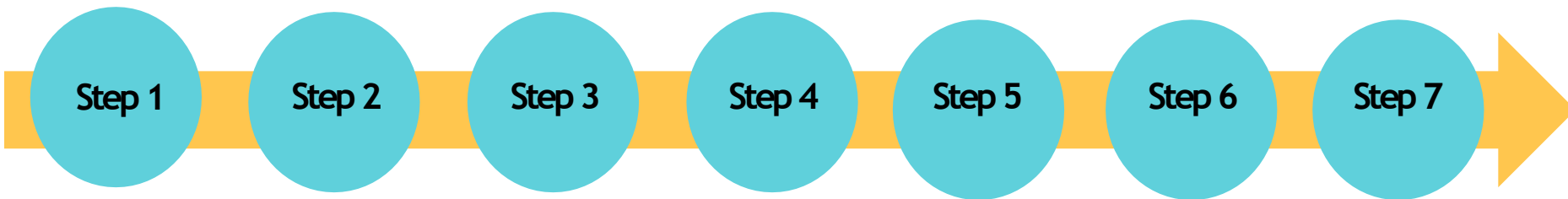
Create a BI Roadmap

Implement (start with some quick wins)



- Executive level sponsorship is key
- Update sponsor regularly
- Quick wins will help

Create a Business Intelligence Strategy



Get Sponsorship

Assemble your BI team

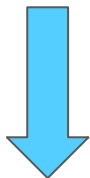
Identify and involve stakeholders

Choose the BI platform and tools

Define the scope

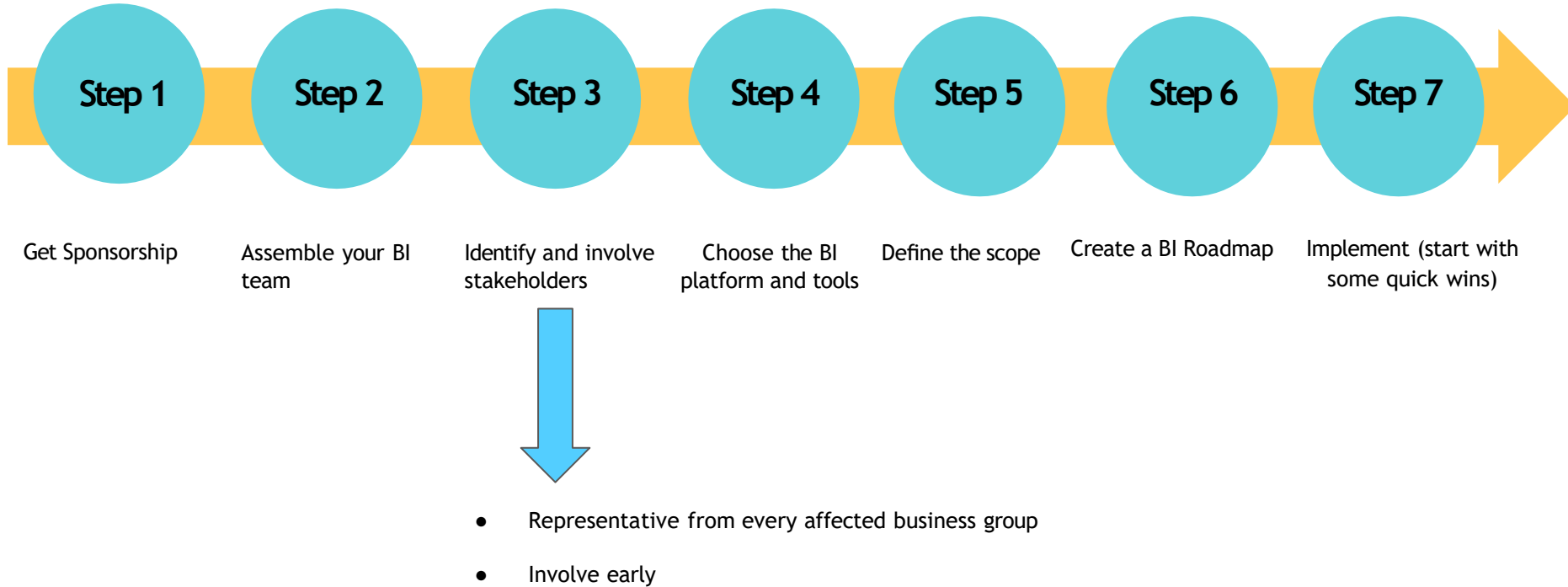
Create a BI Roadmap

Implement (start with some quick wins)

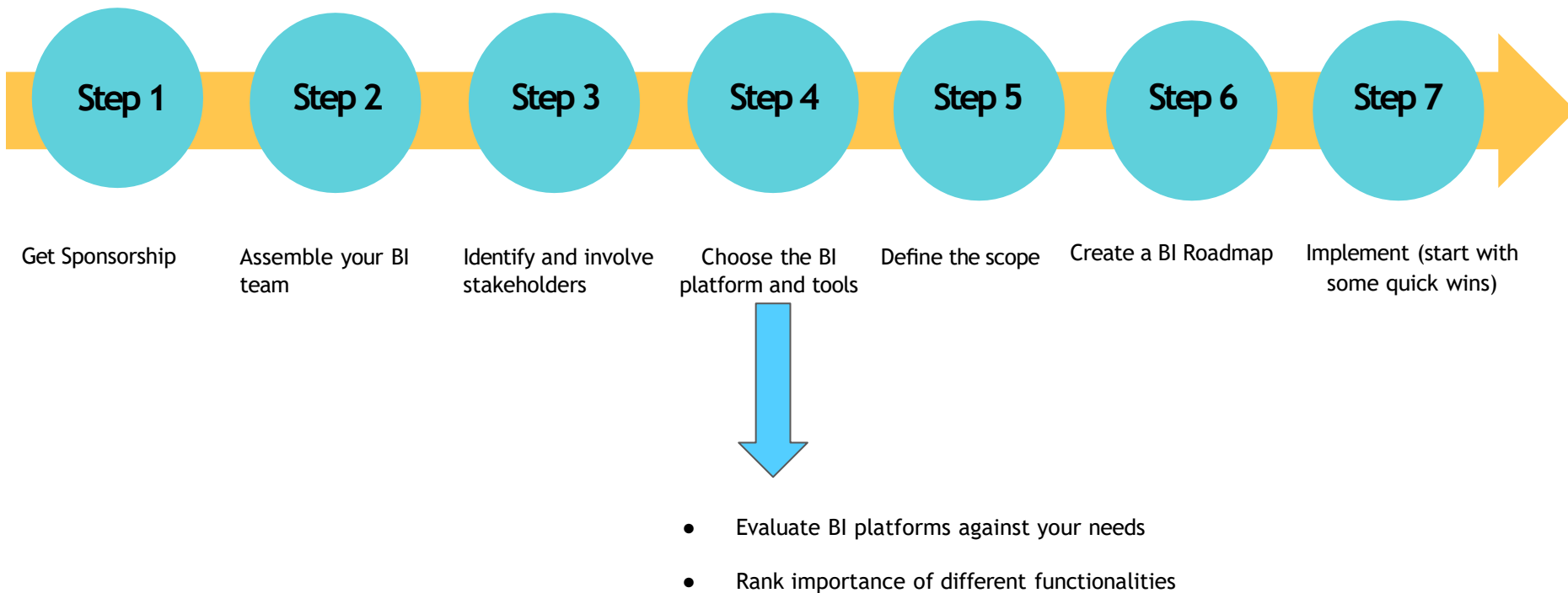


- Program Manager
- IT Owner
- Enterprise Architect
- Data Stewards
- etc

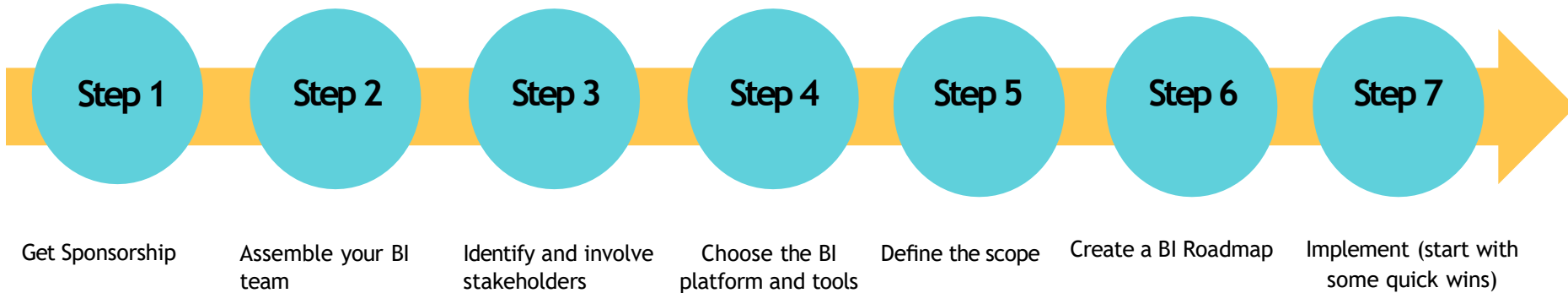
Create a Business Intelligence Strategy



Create a Business Intelligence Strategy

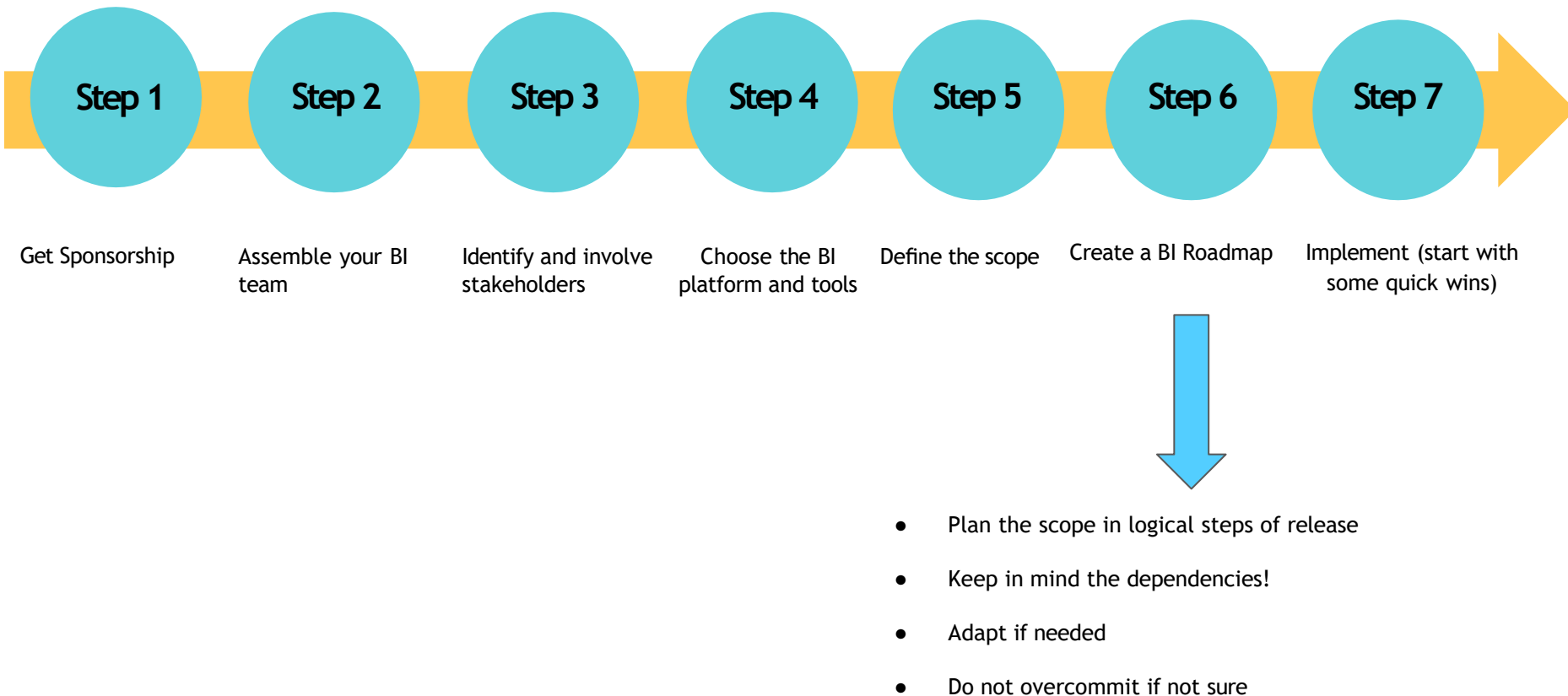


Create a Business Intelligence Strategy

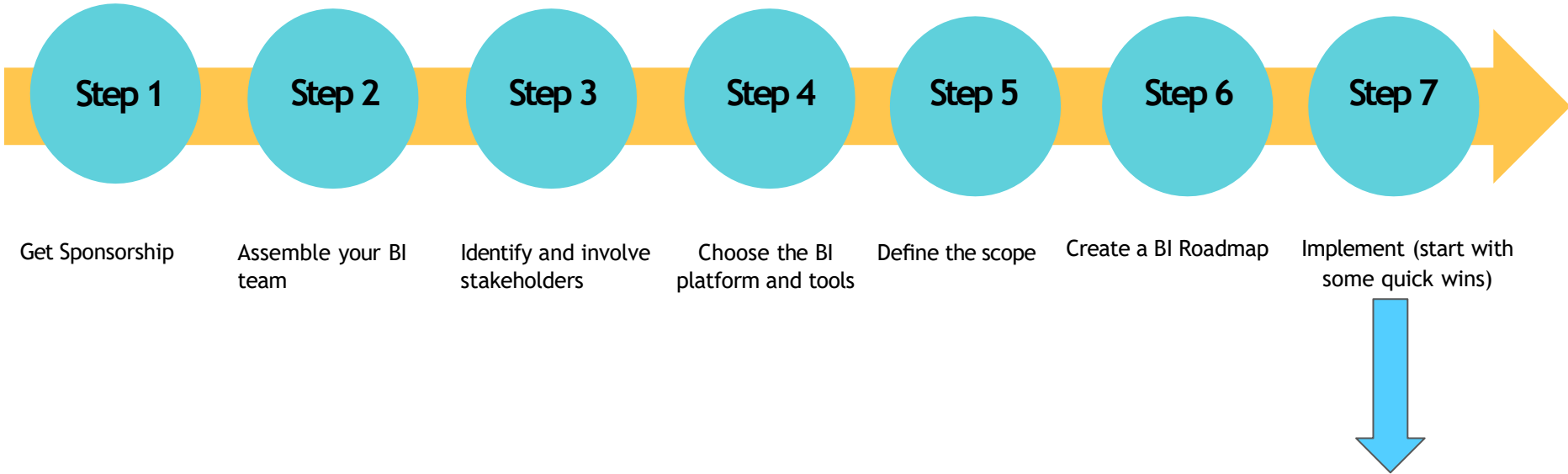


- What do you want to change?
- What do you envision the processes to look like?
- Do you start from a limited scope first?
- What will be the scope of analysis done?
- What is priority for the organization?

Create a Business Intelligence Strategy



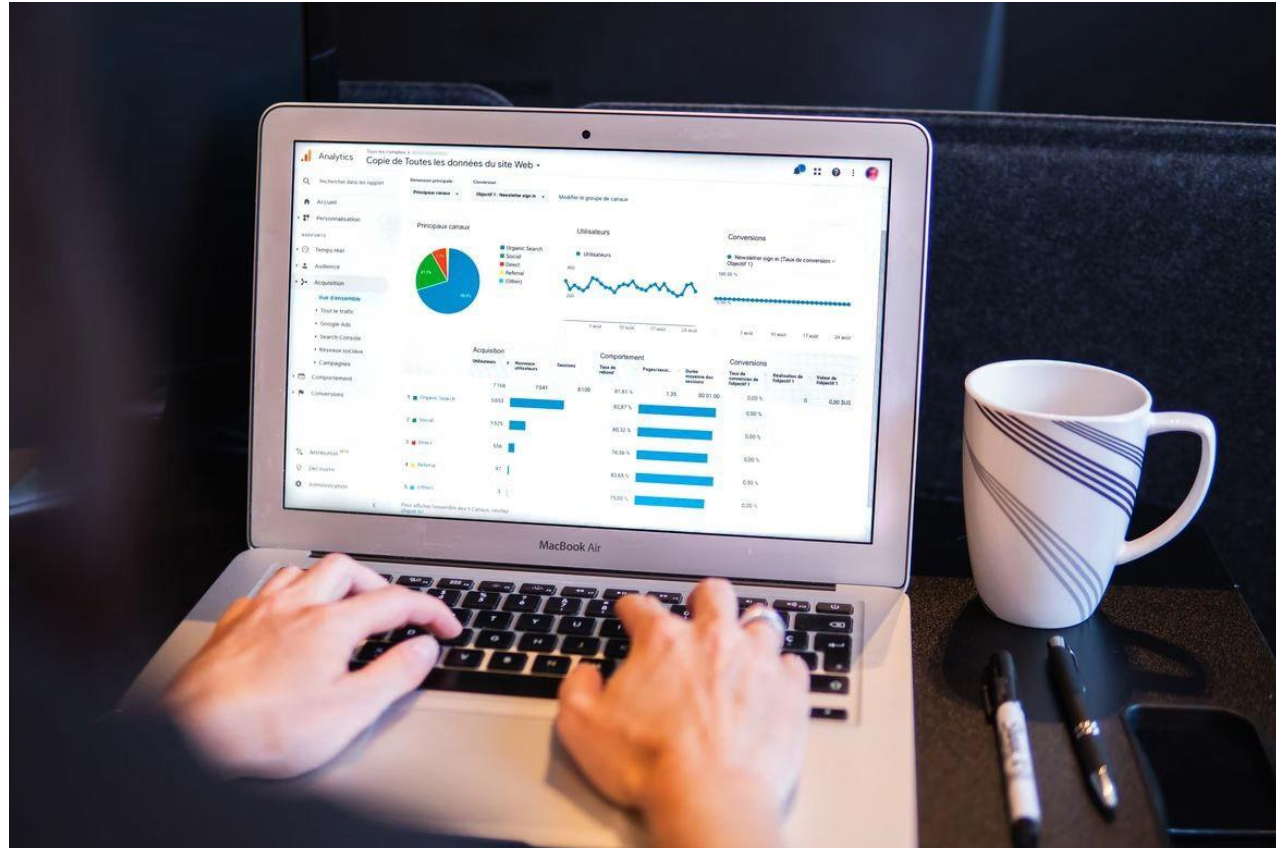
Create a Business Intelligence Strategy



- Have strong partnership between stakeholders, IT and BI teams
- Do not forget about security and security profiles
- Break up implementation into multiple phases
- Do not forget about training and change management

Big Data and BI

- What is Big Data?
- Combining BI and Big Data



Self-Service Business Intelligence

- What is Self-Service BI
- Why is it important
- What are the challenges
- Best Practices



Magic Quadrant for Analytics and BI Platforms

Figure 1: Magic Quadrant for Analytics and Business Intelligence Platforms



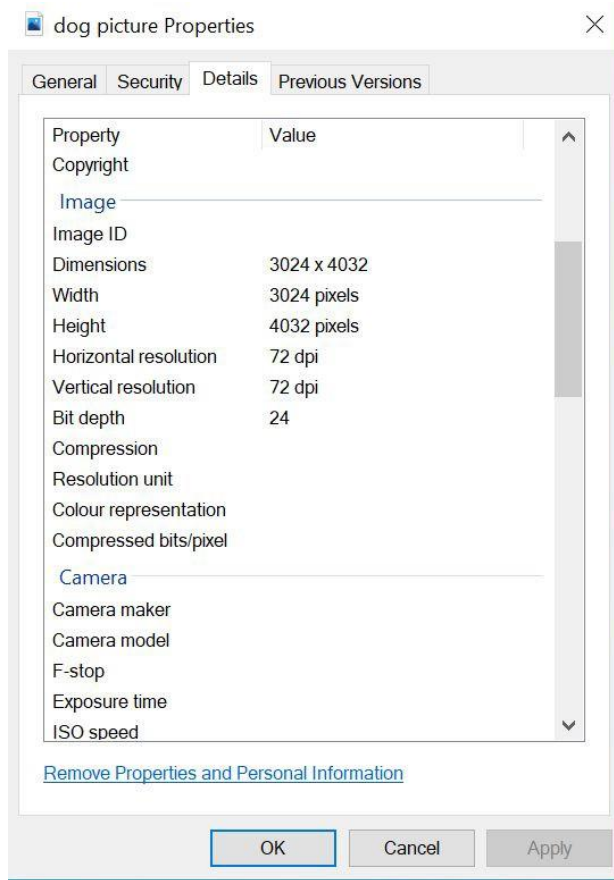
Source: Gartner (March 2022)

Module 12 - Metadata Management

What is Metadata

“**Metadata** is data that provides information about other data”

Example of Metadata



What is Metadata Management?

It is the portfolio of best-practice processes and technologies that allow businesses to manage this data about their data and derive insights for more effective data management. It allows users of all kinds - business, technical, and operational - to search for, understand, and securely access the data they need to do their jobs.

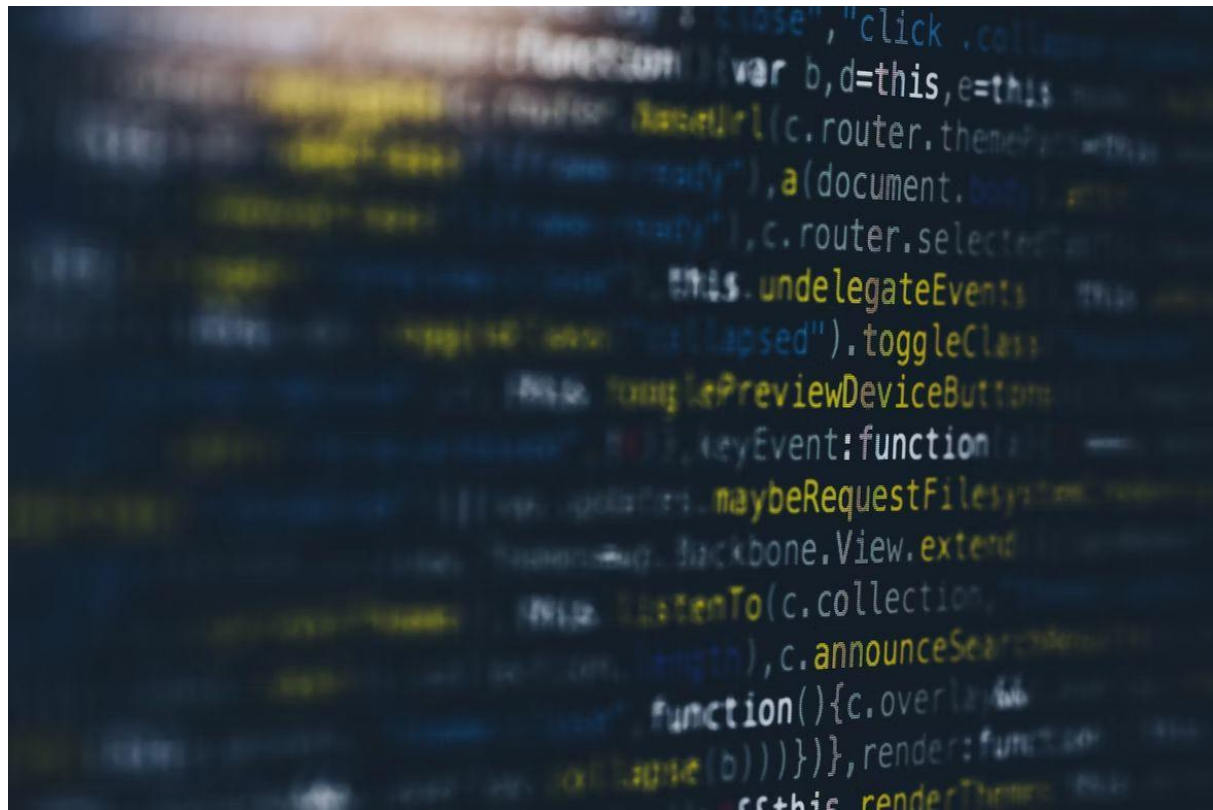
Why do you need Metadata Management?

- Improved Consistency - establish a common business language
- Capture institutional knowledge
- Better data quality
- Faster access to insights
- Faster project delivery timelines
- Reduced costs
- Improved regulatory compliance



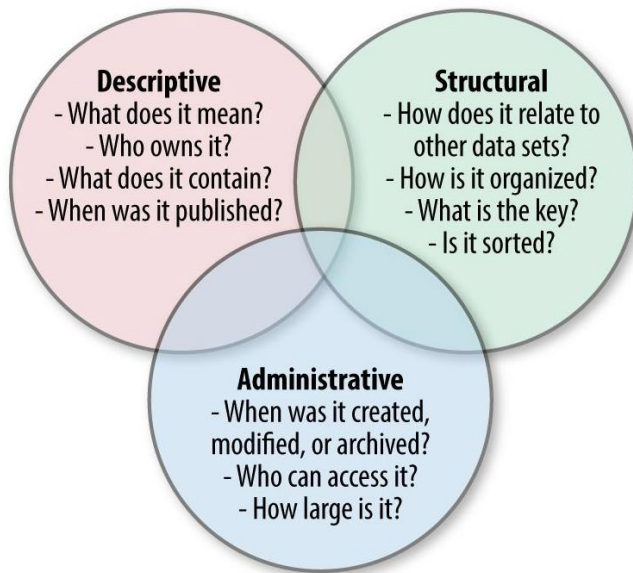
Types of Metadata

- Descriptive metadata
- Structural metadata
- Administrative metadata



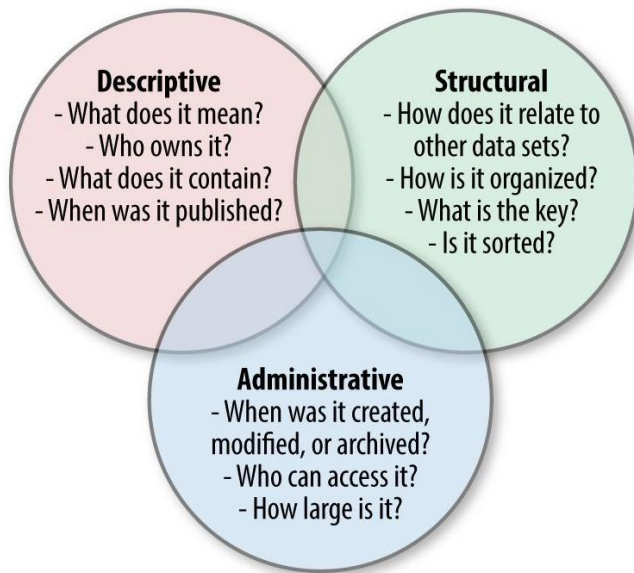
Descriptive Metadata

Descriptive metadata describes a resource for purposes such as discovery and identification. It can include elements such as title, abstract, author, and keywords.



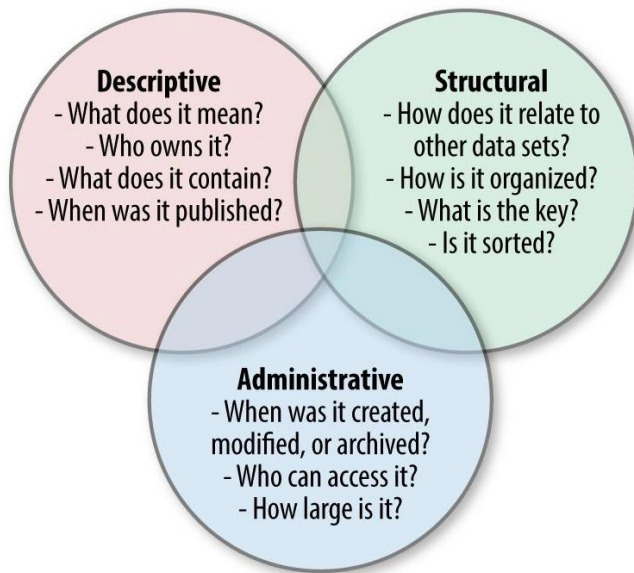
Structural Metadata

Structural metadata is used to specify the relationships between components of a digital object (internal structure) and between different digital objects (external structure)



Administrative Metadata

Administrative metadata provides information to help manage a resource, such as when and how it was created, file type and other technical information, and who can access it.



Metadata Activities

1

Understand Metadata Requirements

2

Define Metadata Strategy

3

Define Metadata Architecture

4

Create and Maintain Metadata

5

Query, Report and Analyze Metadata

UNDERSTANDING METADATA REQUIREMENTS

Explanation: This involves determining what kind of metadata is necessary for the organization's data systems. It's about identifying the types of metadata that are essential for data processing, compliance, and business intelligence.

Example: An IT company developing a cloud storage service needs to understand metadata requirements for file storage. This could include file types, sizes, creation and modification dates, access permissions, and user activity logs. This metadata helps in managing storage systems, providing user access history, and optimizing data retrieval.



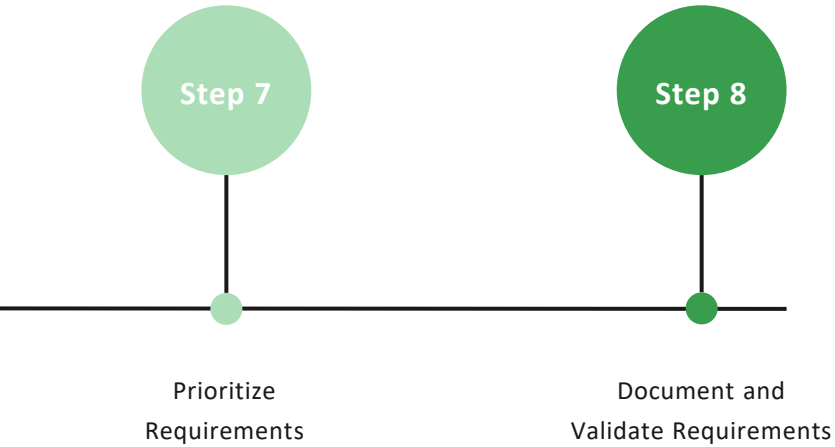
Understand Metadata Requirements

Step by Step



Understand Metadata Requirements

Step by Step



DEFINE METADATA STRATEGY

Explanation: This step is about outlining a plan for how metadata will be handled within the organization. It includes the selection of metadata standards, the definition of processes for metadata creation, update, and deletion, and the establishment of roles for metadata governance.

Example: The IT department of a financial institution might define a metadata strategy that includes using the ISO 20022 standard for metadata concerning financial transactions. The strategy would detail how metadata is to be captured at each transaction point and the process for updating it to reflect transaction status changes.



DEFINE METADATA ARCHITECTURE

Explanation: This activity involves designing the technical framework that supports the collection, storage, and dissemination of metadata. It includes the selection of tools and technologies, the design of metadata models, and the integration of metadata with other data management systems.

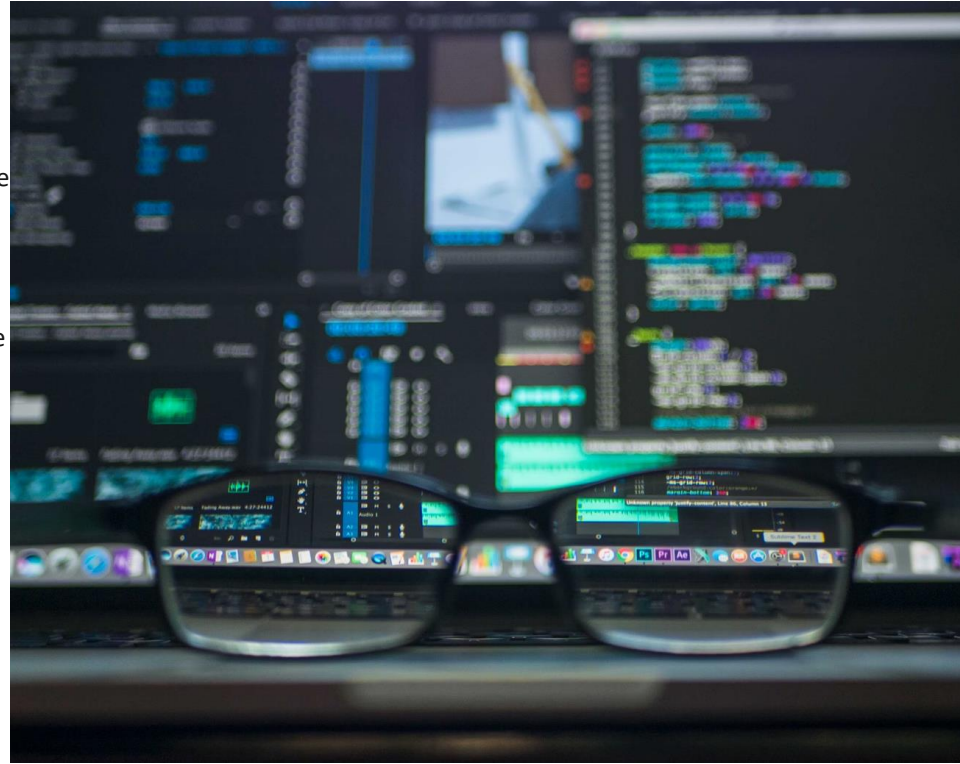
Example: A software development company might design a metadata architecture that includes a centralized metadata repository, which integrates with their version control system. This would allow for tracking changes in code, dependencies, and deployment configurations, facilitating better software lifecycle management.



CREATE AND MAINTAIN METADATA

Explanation: This is the operational phase where metadata is generated according to the established requirements and strategy, and it is kept up-to-date to reflect the current state of the underlying data.

Example: An IT service provider could create metadata for each of its services, including service name, version, dependencies, configurations, and performance metrics. Maintenance would involve updating the metadata when services are upgraded, dependencies are changed, or new configurations are added.



QUERY, REPORT, AND ANALYZE METADATA

Explanation: This involves using metadata to gain insights into the IT infrastructure, improve data governance, and support decision-making. It includes building tools to search metadata, generate reports, and conduct analyses.

Example: An IT company might use metadata queries to monitor the performance of its network infrastructure. By analyzing metadata on network traffic, device status, and incident reports, the company can identify bottlenecks, predict hardware failures, and optimize network performance.



Magic Quadrant for Metadata Management Solutions

Figure 1. Magic Quadrant for Metadata Management Solutions



Source: Gartner (November 2020)

Implement Metadata Management

Step 1



Select the Metadata
Program team

Step 2



Define the
Metadata Strategy

Step 3



Adopt Metadata
Standards

Step 4



Get the right
Metadata
Management tool

Step 5



Deploy and expand



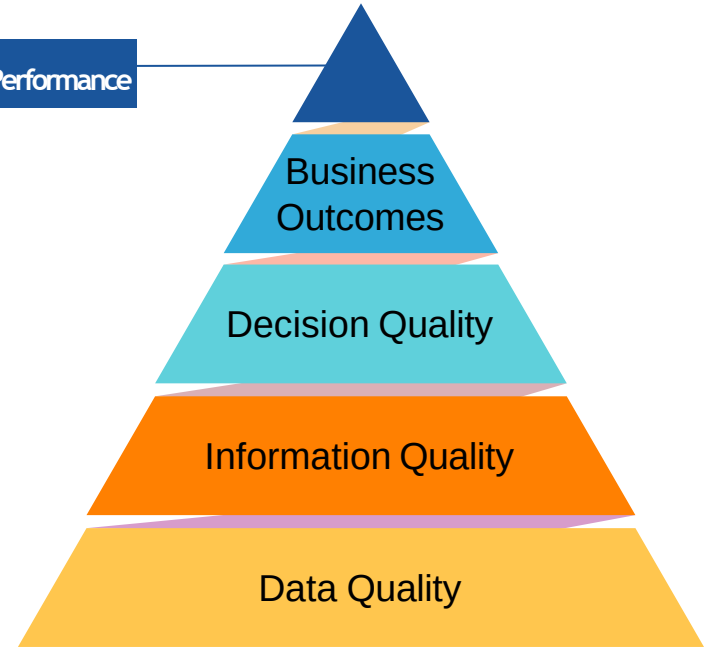
Module 13 - Data Quality Management

What is Data Quality?

Simple Definition of Data Quality:

"Data quality is defined by how well a given dataset meets a user's needs. Data quality is an important criteria for ensuring that data-driven decisions are made as accurately as possible"

Company Performance



Data Quality Management

Definition of Data Quality Management:

"Set of practices that aim at improving and maintaining a high quality of information within the organization"

Pillars of Data Quality Management

People

Data Profiling

**Defining Data
Quality**

Data reporting

Data Repair

Cost of Poor Data Quality

- According to IBM's estimate, the US lost \$3.1 trillion yearly due to bad data.
- Gartner.com suggests that organizations lose between \$10 to \$14 Million USD annually due to poor data.
- Integrate reported that around 40% of all leads have inaccurate data.
- Cio.com identified that around 80% of companies believe they lost revenue due to data challenges.
- MIT Sloan reported that employees spend half of their time coping with managing data quality tasks.
- Pragmaticworks states 20 to 30 percent of operating expenses are due to bad data.
- Econsultancy.com reported that due to poor data, companies having mail delivery issues lost about 30% of their revenue, in addition to the 21% of businesses experienced reputation damages.
- Gartner also reported that data scientists spend around 80% of their time cleaning and organizing data.

Data Quality Dimensions



Data Quality Improvement Process

Step 1



Define the Data
Quality improvement
goals

Step 2



Data Profiling

Step 3



Conduct Data
Quality
Assessment

Step 4



Resolve Data
Quality Issues

Step 5



Monitor and
Control



Module 14 - Big Data & Data Science

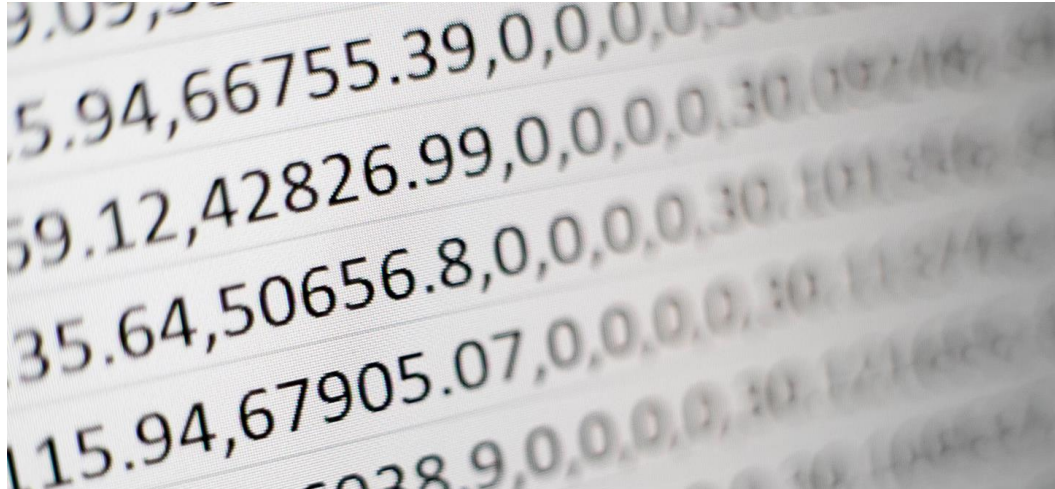
Definition of Big Data

Big data refers to extremely large datasets that are so complex and vast that traditional data processing software cannot handle efficiently.

Key Characteristics:

- **Large Volumes:** Think of the massive amounts of information generated every second, like social media posts, online transactions, and sensor data from devices.
- **Fast Processing:** The data is generated quickly, often in real time, requiring immediate analysis to make timely decisions.
- **Different Types:** Big data comes in various forms, including text, images, videos, and more, making it challenging to analyze.

In simple terms, big data is about managing and analyzing huge amounts of information to find patterns and insights that can help businesses and organizations make better decisions.



Uber and Big Data

Here are 5 primary applications of big data at Uber:

- Dynamic Pricing and Surge Pricing
- Driver-Rider Matching
- Predictive Analytics for Demand Forecasting
- Route Optimization
- Fraud Detection and Safety Measures

<https://d3.harvard.edu/platform-digit/submission/uber-knows-you-how-data-optimizes-our-rides/>

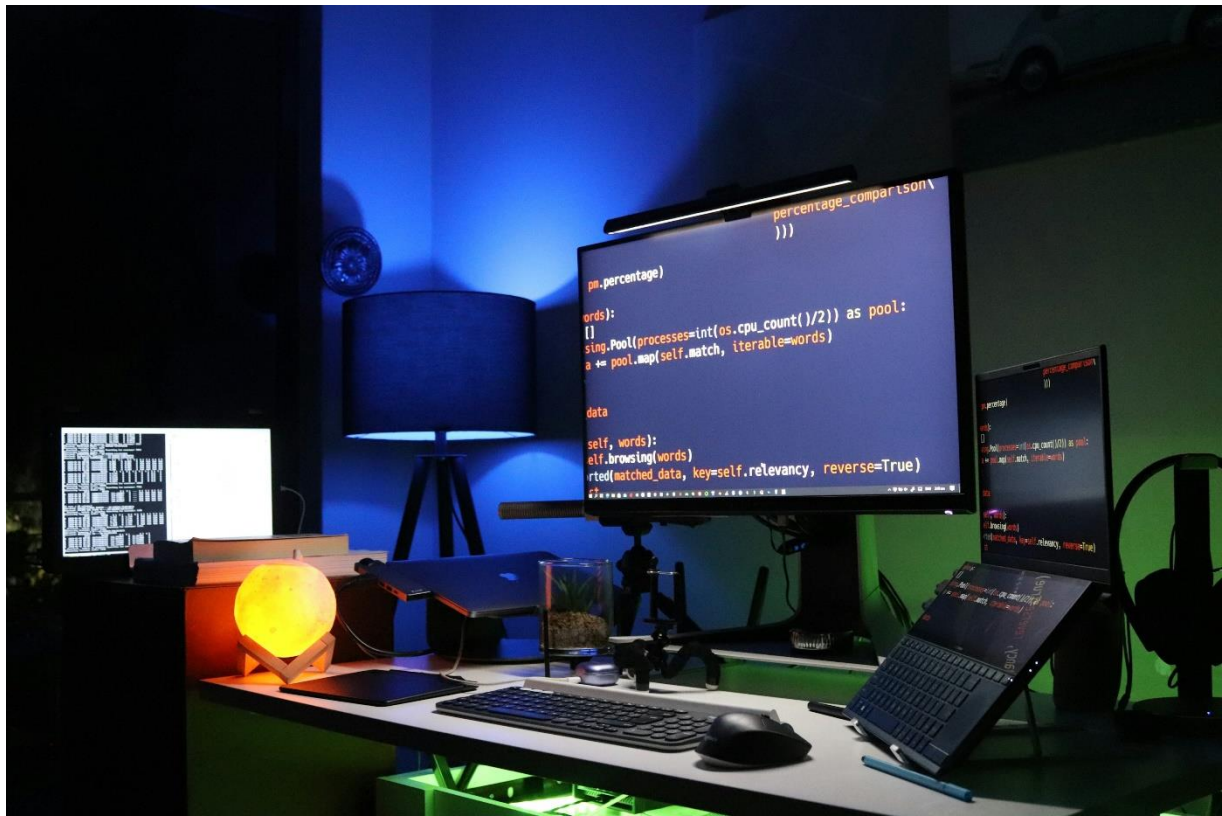


Definition of Data Science

Data science is a field that combines various techniques and theories from statistics, mathematics, computer science, and domain knowledge to extract meaningful insights from structured and unstructured data. It involves processes such as data collection, cleaning, analysis, and visualization to inform decision-making and drive business strategies.

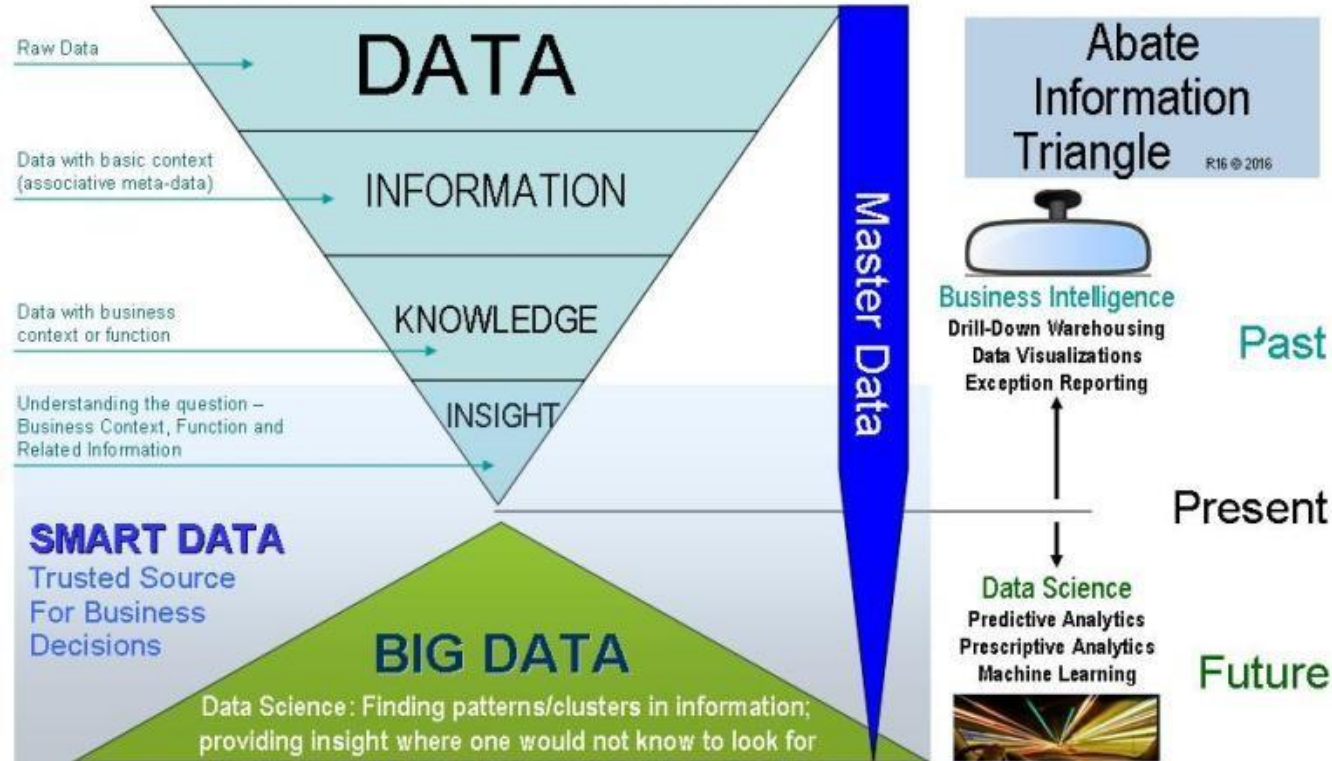
Key skills in Data Science:

1. Statistical Analysis
2. Programming
3. Machine Learning
4. Data Visualization

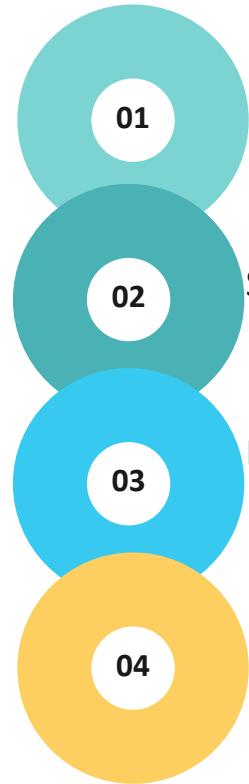


The Abate Information Triangle

The Abate Information Triangle is a foundational model developed by Robert J. Abate that illustrates the relationship between Data, Information, and Knowledge. This framework has gained particular relevance in the context of Big Data, highlighting how organizations can effectively manage and utilize vast amounts of information.



GOALS OF BIG DATA AND DATA SCIENCE

- 
- 01 Discover relationships between data and the business
 - 02 Support the iterative integration of data sources(s) into the enterprise
 - 03 Discover and analyze new factors that might affect the business
 - 04 Publish data using visualization techniques in an appropriate, trusted and ethical manner.

What Big Data actually involves

7 Key Activities

1. Define Big Data Strategy and Business Needs

2. Choose Data Sources

3. Acquire and Ingest Data Sources

4. Develop Hypotheses and Methods

5. Integrate data for analysis

6. Explore Data Using Models

7. Monitor and Deploy

Deliverables of Big Data

6 Key Deliverables

1. Big Data Strategy and Standards

2. Data Sourcing Plan

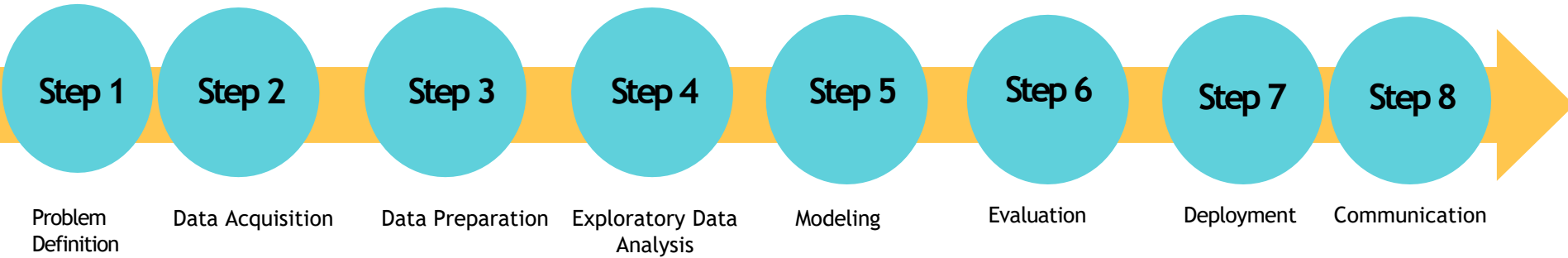
3. Acquired Data Sources

4. Initial data analysis and hypotheses

5. Data insights and findings

6. Enhancement Plan

The Data Science Lifecycle



Step 1 - Summarize the data

- We summarize to simplify the data
- We summarize to quickly identify “normal” and “odd”
- We summarize to provide more context to the data



Four key areas of data summarization

- Centrality
- Dispersion
- Replication (aggregation)
- Shape



Data Centrality

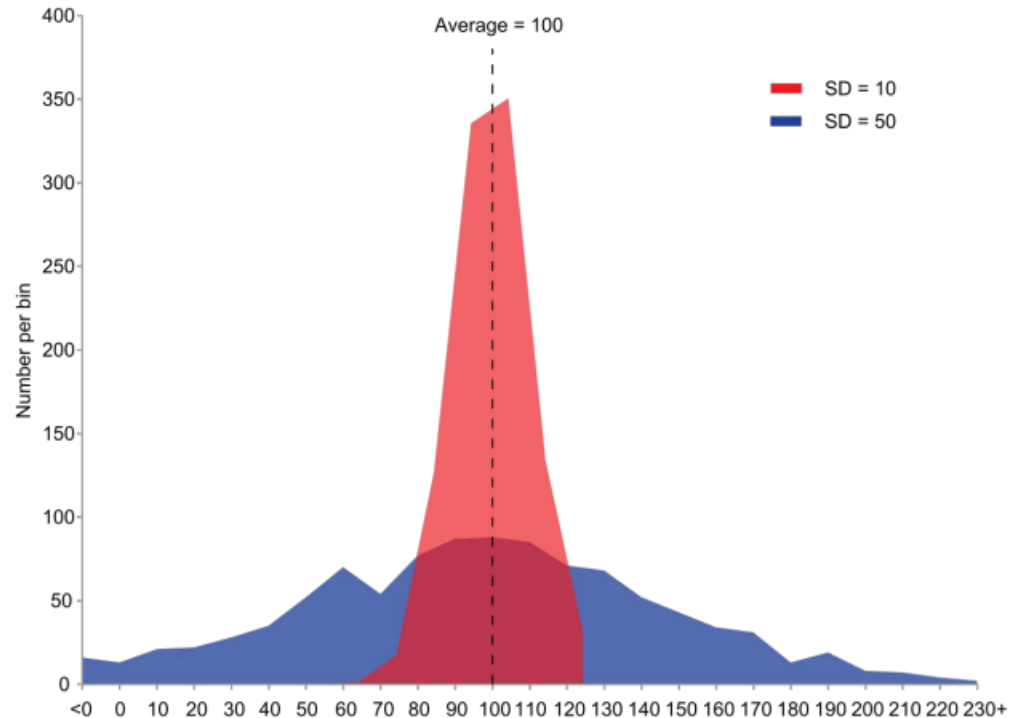
$$\text{Mean } \bar{x} = \frac{\sum x_i}{N}$$

$$\text{Median} = \begin{cases} \frac{(N+1)^{\text{th}}}{2} \text{ term; when } N \text{ is odd} \\ \frac{\frac{N}{2} \text{ term} + \left(\frac{N}{2} + 1\right) \text{ term}}{2} ; \text{ when } N \text{ is even} \end{cases}$$

Mode = The value in the data set that occurs most frequently

Data Dispersion

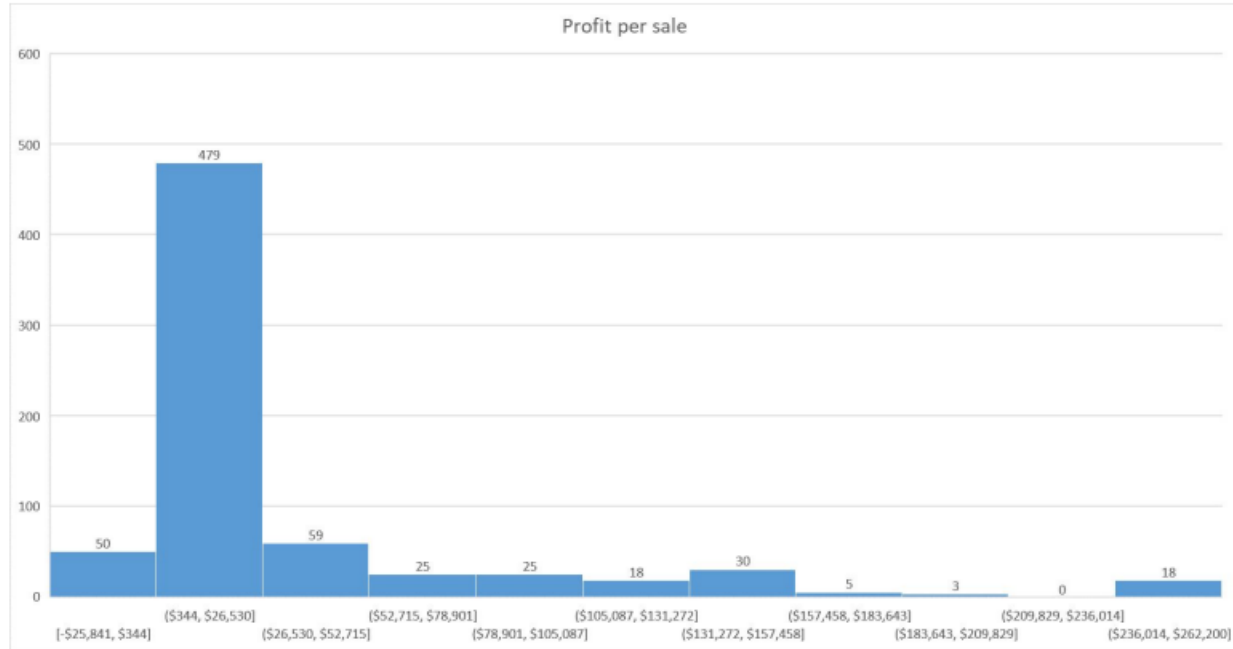
- Range
- Standard Deviation
- Inter-Quartile range



Data Repetition

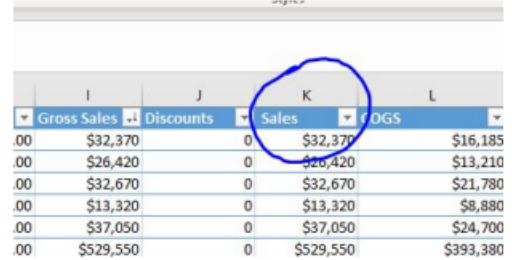
	A	B	C	D	E	F	G	H
1	Year	2023						
2								
3	Sum of Sales	Column Labels						
4	Row Labels	Hat	Jeans	Shirt	Shoes	Socks	T-Shirt	Grand Total
5	France	\$2,943,703	\$1,005,746	\$2,101,032	\$528,537	\$4,187,817	\$582,703	\$11,349,538
6	Germany	\$2,158,598	\$2,983,373	\$6,498,114	\$5,671,875	\$6,434,146	\$327,027	\$24,073,134
7	India	\$4,148,103	\$1,279,961	\$1,005,718	\$3,593,731	\$870,321	\$115,116	\$11,012,950
8	United Kingdom	\$2,521,553	\$1,162,950	\$45,540	\$1,697,265	\$820,275	\$554,514	\$6,802,097
9	United States of America	\$2,942,177	\$5,306,425	\$798,099	\$1,040,857	\$2,225,964	\$136,749	\$12,450,272
10	Grand Total	\$14,714,134	\$11,738,455	\$10,448,504	\$12,532,265	\$14,538,523	\$1,716,110	\$65,687,990
11								

Data Shape



Exercise - Histogram

1. Create a Histogram out of the Sales column
2. Copy the Histogram into a new worksheet called "Histogram"
3. Change the Title to Order Sales \$ Frequency
4. Change the bin sizes to 100,000 width
5. Change the bars to green color
6. Add data labels



The screenshot shows an Excel spreadsheet with columns labeled I, J, K, and L. The data is as follows:

	I	J	K	L
	Gross Sales	Discounts	Sales	COGS
.00	\$32,370	0	\$32,370	\$16,185
.00	\$26,420	0	\$26,420	\$13,210
.00	\$32,670	0	\$32,670	\$21,780
.00	\$13,320	0	\$13,320	\$8,880
.00	\$37,050	0	\$37,050	\$24,700
.00	\$529,550	0	\$529,550	\$393,380

Exercise - Pivot Chart

1. Insert a Pivot Chart from the Cali Fashion data
2. Name the new worksheet "Pivot Chart"
3. The value field you will be using is **Sales**
4. Your axis will be **Country** followed by **Segment**
5. Add a title - "Sales per Country and Segment"
6. Remove the axis buttons from the bottom of the chart
7. Remove Sum of Sales button and Total label
8. Add label to the bars and format to currency with no decimals
9. Change the color of the bars
10. Play around with the drill-down functionality



Congratulations!